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Chapter 12

GELFAND-P000738

[source number: 6]

21/XI-77. Gelfand Sem.



Diagram and typed description.
Yu. I. Manin.

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(on work with V. Drinfeld)

Description of self-dual solutions of the Yang–Mills eqs.

$\mathbb{R}^4 \ni (x_1, x_2, x_3, x_4)$. Unknown functions:

$$A_\mu(x) \in u(r), \qquad r \geq 2$$

(Hermitian or anti-Hermitian matrices; in the Lie algebra)

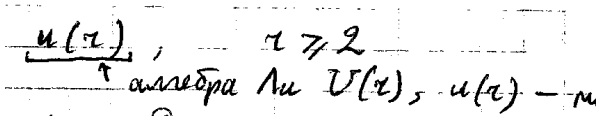


Diagram and typed description.
$\underbrace{u(r)}$
Lie algebra of $U(r)$

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$u(r)$ — the set of skew-Hermitian $r \times r$ matrices.

$$F_{\mu\nu}(x) \underset{\text{def}}{=} \frac{\partial}{\partial x_\mu} A_\nu - \frac{\partial}{\partial x_\nu} A_\mu + [A_\mu, A_\nu] \tag{1}$$

1) Duality equations:

$\rho\sigma$ — complementary pair of indices to $\mu\nu$, $F_{\mu\nu} = \pm F_{\rho\sigma}$; more precisely:

$$F_{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} \tag{2}$$

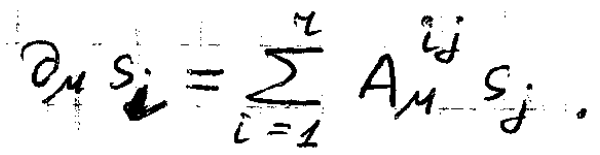
($\varepsilon^{\mu\nu\rho\sigma}$ — the sign of the permutation).

2) Regularity conditions: A) at finite points B) at infinity

A) singularities at finite points can be removed by a gauge transformation.

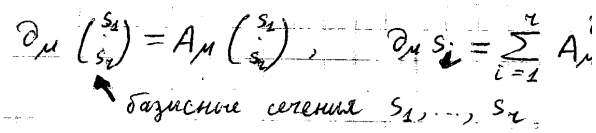
More precisely, consid. the bundle $\mathbb{C}^r \times \mathbb{R}^4$ and a basis of sections in it; the covariant differential, determined by the matrices A_μ, is given by the formula:

$$\partial_\mu \begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix} = A_\mu \begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix}, \qquad \partial_\mu s_i = \sum_{i=1}^{r} A_\mu^{ij} s_j. \tag{3}$$



Uncertain reading. The lower limit of summation reads $i = 1$. The i in s_i on the left is heavily retraced.

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Diagram and typed description.

$$\partial_\mu \begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix}$$

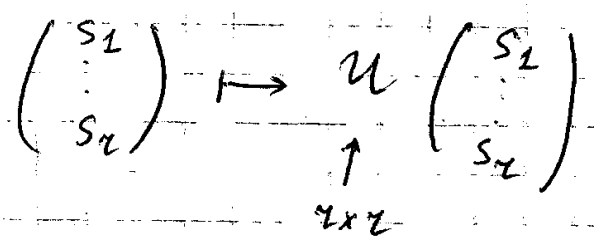
basis sections $s_1, \dots, s_r$

Pass to another basis. We obtain a transformation (gauge transformation).

$$A_\mu \mapsto u^{-1}\partial_\mu u + u^{-1}A_\mu u = A'_\mu \tag{4}$$

Regularity condition at a finite point: there is a genuine connection without singularities, i.e. the sing. in A_μ can be removed by a gauge transformation.

Transition:



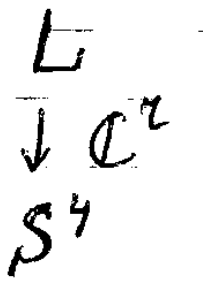
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Diagram and typed description.

$$\begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix} \mapsto \underset{r \times r}{u} \begin{pmatrix} s_1 \\ \vdots \\ s_r \end{pmatrix}$$

B) Conditions at $P = \infty$.

The bundle on $\mathbb{R}^4$ extends to a bund. over $S^4 = \mathbb{R}^4 \cup \{\infty\}$.



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Diagram and typed description.

$$\begin{array}{c} L \\ \downarrow \mathbb{C}^r \\ S^4 \end{array}$$

L — the space of our bundle.

Regularity cond.: The connection ∂_μ extends to a connection in L over S^4 (the bundle L need no longer be trivial)

This also means that a twist at infinity can remove the singularity.

An essential invariant of the solution is the bundle L. This is a discrete invariant. On L there is a unitary bundle structure.

Classification of L.

Gluing the trivializations on the equator gives a map $S^3 \to U(r)$, i.e. an element of $\pi_3(U(r)) = \mathbb{Z}$.

($SU(2) \subset U(r)$, $SU(2) \cong S^3$; hence an integer-valued sufficient invariant = degree of the map n = Pontryagin number)



Uncertain reading. The word after "integer-valued" reads "достаточный" (sufficient). Its first letters are heavily retraced.

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the embedding $SU(2) \subset U(r)$ is taken in the form

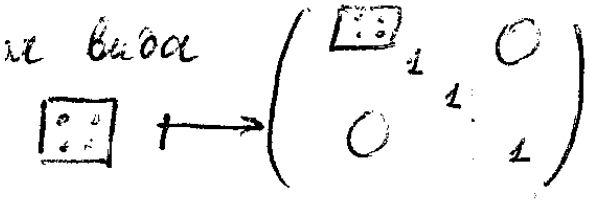


Diagram and typed description.

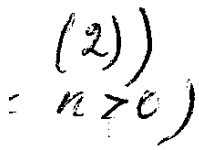
$$\square \longmapsto \begin{pmatrix} \square & 0 \\ 1\ 1 & \\ 0 & 1 \end{pmatrix}$$

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We shall assume $n \geq 0$ and the duality conditions

($n < 0 \Rightarrow$ we can change the sign in eq. (2))

— anti-self-dual solns. — reduce to $n > 0$)

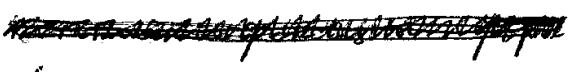


Uncertain reading. Closing parentheses are visible after "(2)" and at the end of the following line.

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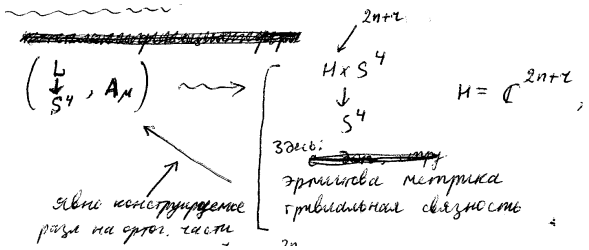
Plan for what follows:

[Crossed-out line]



Undeciphered passage. The line below the heading is heavily crossed out and illegible.

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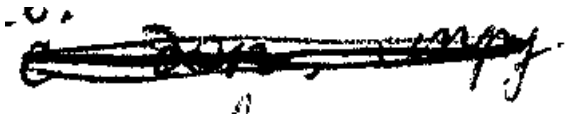
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Diagram and typed description.

$$\begin{pmatrix} L \\ \downarrow \\ S^4 \end{pmatrix}, A_\mu \rightsquigarrow \left[\begin{array}{l} \overset{2n+r}{\swarrow} \\ H \times S^4 \\ \downarrow \\ S^4 \end{array}\right. \qquad H = \mathbb{C}^{2n+r},$$

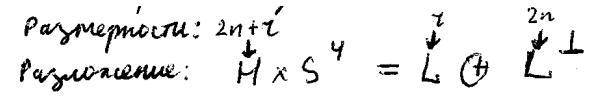
Here:
[crossed out]
Hermitian metric
trivial connection

Explicitly constructed
decomp. into orthog. parts



Original source details: GELFAND-P000739

Undeciphered passage. A short phrase is crossed out inside the large bracket. The first letter resembles Cyrillic “c”.



Original source details: GELFAND-P000739

Diagram and typed description.

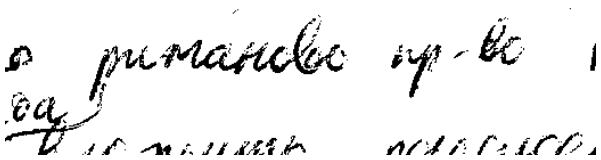
$$\begin{array}{lccccc} \text{Dimensions:} & 2n+r & & r & & 2n \\ & \downarrow & & \downarrow & & \downarrow \\ \text{Decomposition:} & H \times S^4 & = & L & \oplus & L^\perp \end{array}$$

$\{A_\mu\}$ — a direct summand of the trivial connection in $H \times S^4$

(Differentiate and project onto L).

Unitarity of the connection is proved.

[This generalizes the fact that a Riemannian space can be embedded in Euclidean space; can we [inserted above: always] embed a bundle with connection?]



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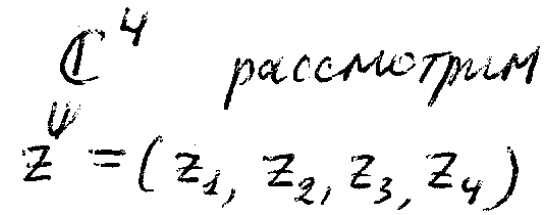
Uncertain reading. The abbreviation after “Riemannian” reads “пр-во” (space).

Which decompositions give self-dual fields?

We must pass from the base S^4 to its complexification.

In $\mathbb{C}^4$ consider the map

$$\sigma : \; z \mapsto (-\overline{z}_2, \overline{z}_1, -\overline{z}_4, \overline{z}_3).$$



Original source details: GELFAND-P000739

Diagram and typed description.

$$\begin{array}{l} \mathbb{C}^4 \\ \downarrow \\ z = (z_1, z_2, z_3, z_4) \end{array}$$

(in quaternions, multiplication by i)

For σ there are no fixed pts. (except 0) and no fixed lines, but there are invariant planes

$$\mathbb{C}z + \mathbb{C}\sigma(z) \subset \mathbb{C}^4 \tag{5}$$

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[source number: 7]

We shall call them real (or σ-invariant) (σ — involution).

$$\{\sigma\text{-invar. planes}\} = S^4 \tag{6}$$

(proof omitted)

(in $\mathbb{C}P^3$ this is the set of σ-invar. lines).

$E^4 \subset$ [crossed out] $S^4 = E^4 \cup \{\infty\}$.



Undeciphered passage. A mark between the inclusion sign and S^4 is heavily crossed out.

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Hence there is a map

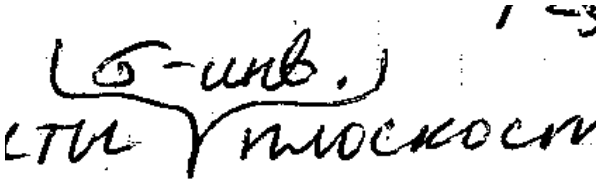
$$\mathbb{C}^4 \setminus \{0\} \xrightarrow{D} S^4 \supset E^4. \tag{7}$$

$$z \longmapsto \sigma\text{-invar. plane through } z.$$

Write D in coordinates (in E^4 the coordinates are (x_i)). The normalization is as in Zakharov–Belavin.

$$(z_1, z_2, z_3, z_4) \longmapsto \begin{cases} x_2 + ix_1 = \dfrac{z_1\bar{z}_3 + \bar{z}_2 z_4}{|z_3|^2 + |z_4|^2}, \\ x_4 + ix_3 = \dfrac{\bar{z}_2 z_3 - z_1\bar{z}_4}{|z_3|^2 + |z_4|^2}. \end{cases}$$

(Plücker coordinates of the [inserted above: σ-inv.] plane through pt. z)



Uncertain reading. The abbreviation "σ-инв." is inserted above the line with a mark indicating its position.

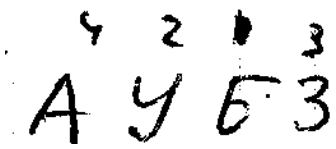
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This is the map

$$\mathbb{C}^4 \setminus \{z_3 = z_4 = 0\} \longrightarrow E^4.$$

Remark: to recover a σ-inv. plane, 3 coordinates suffice.

Lemma 1 ($^{4}_{\text{A}}\,^{2}_{\text{W}}\,^{1?}_{\text{B}}\,^{3}_{\text{Z}}$ — Atiyah, Ward, Belavin, Zakharov).



Uncertain reading. Above "Б" is a heavy vertical mark, tentatively 1. The other numbers read 4, 2 and 3.

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Uncertain reading. The beginning of “Belavin” is heavily retraced.

a)

$$D^*(dx_\mu \wedge dx_\nu + \varepsilon^{\mu\nu\rho\sigma} dx_\rho \wedge dx_\sigma) \quad \text{— a form of type } (1,1).$$

(Consider such forms and lift them to $\mathbb{C}^4 \setminus 0$ using D)

b)

$$\left[D^* \begin{pmatrix} dx_1 \wedge dx_2, \\ dx_1 \wedge dx_3, \\ dx_1 \wedge dx_4 \end{pmatrix} \right]^{2,0} \quad \text{linearly independent.}$$

[crossed out]



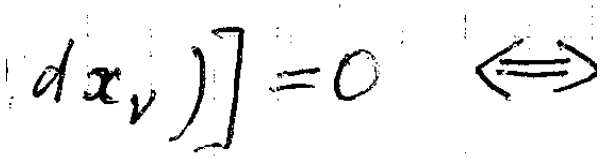
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Undeciphered passage. A short fragment after “linearly independent” is heavily crossed out.

$(2,0)$-components.

Corollary 1 (AW).

$$\left[D^* \left(\sum F_{\mu\nu}\, dx_\mu \wedge dx_\nu \right) \right] = 0 \quad \Longleftrightarrow \quad F_{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}. \tag{8}$$

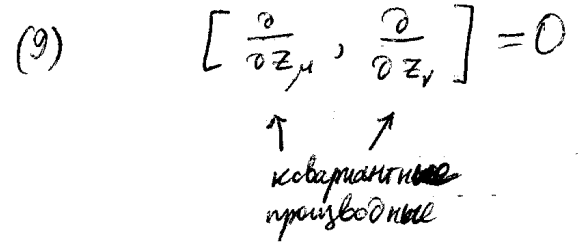


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Uncertain reading. The closing square bracket in (8) has no written component index.

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Corollary 1′ (BZ)



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Diagram and typed description.

$$(9) \qquad \left[\frac{\partial}{\partial z_\mu}, \frac{\partial}{\partial z_\nu} \right] = 0$$

↑ ↑

covariant derivatives



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Uncertain reading. The ending of the Russian word for “covariant” is heavily retraced.

Belavin. Or:

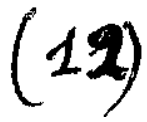
$$[\nabla_\mu, \nabla_\nu] = 0, \tag{10}$$

where

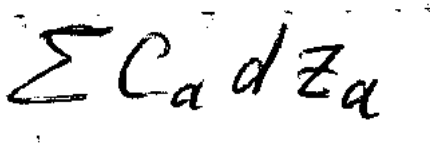
$$\nabla_\mu = \frac{\partial}{\partial z_\mu} + C_\mu, \tag{11}$$

$$\left[D^*\left(\sum A_\mu\, dx_\mu\right)\right]^{1,0} = \sum C_a\, dz_a \tag{12}$$

Uncertain reading. The digit 2 in equation number (12) is heavily retraced.



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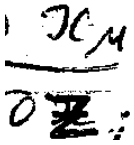
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Uncertain reading. The indices on the right of (12) tentatively read a, possibly α.

Or: set

$$C_i(z) = \frac{\partial x_\mu}{\partial z_i} A_\mu(x) \tag{13}$$

Uncertain reading. The z in the denominator of (13) is heavily retraced.



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Then

[crossed out] $$\frac{\partial C_k}{\partial z_i} - \frac{\partial C_i}{\partial z_k} + [C_i, C_k] = 0 \tag{14}$$

Undeciphered passage. The first sign of the crossed-out group resembles ∂. The other signs are illegible.



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by virtue of the duality equations.

Or:

$$[L_1, L_2] = 0, \tag{15}$$

where

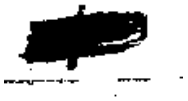
$$L_1 = \lambda(\nabla_1 - i\nabla_2) + (\nabla_3 + i\nabla_4), \tag{16′}$$

Uncertain reading. The sign after the minus in (16′) is heavily retraced and tentatively reads i.



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$$L_2 = \lambda(\nabla_3 - i\nabla_4) - (\nabla_1 + i\nabla_2), \tag{16″}$$

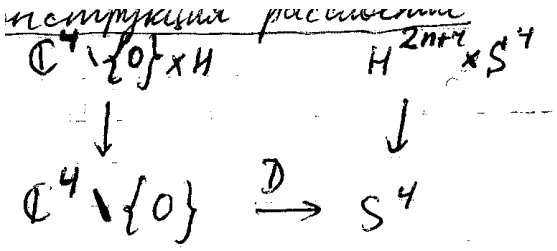


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Uncertain reading. Between the parentheses in (16″) there is a heavy horizontal stroke and short vertical traces. It is tentatively read as a minus.

$$\nabla_\mu \equiv \partial_\mu + A_\mu \tag{17}$$

M. Construction of bundles. They will be constructed on the left.



Original source details: GELFAND-P000741

Diagram and typed description.

$$\begin{array}{ccc} \mathbb{C}^4 \setminus \{0\} \times H & & H^{2n+r} \times S^4 \\ \downarrow & & \downarrow \\ \mathbb{C}^4 \setminus \{0\} & \xrightarrow{\quad D \quad} & S^4 \end{array}$$

Objects of linear algebra $\Rightarrow$ direct summand in $H \times S^4$.

H — $(2n + r)$-dimensional Hermitian space,
form Q.

Consider [crossed out]

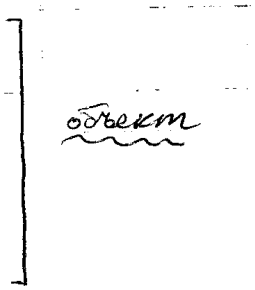
$$I \subset H \otimes_{\mathbb{C}} \mathbb{C}^4 \qquad (*) \tag{18}$$



Original source details: GELFAND-P000741

Undeciphered passage. A separate small crossed-out mark appears before (18).

I — n-dimensional subspace



Original source details: GELFAND-P000741

Diagram and typed description.

object

The right bracket covers the passage from the definition of H to the line about the subspace I.

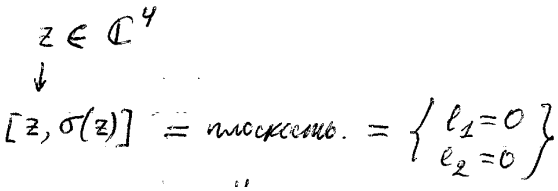
GELFAND-P000742

[source number: 8]

Conditions.

From the object we construct something like a bundle.

$L(*) =$ “system of linear spaces over S^4”.

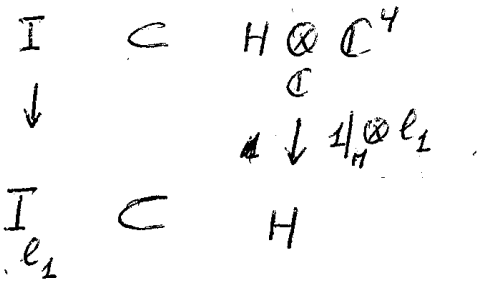


Original source details: GELFAND-P000742

Diagram and typed description.

$$z \in \mathbb{C}^4 \;\downarrow\; [z, \sigma(z)] \quad = \text{plane.} = \left\{ \begin{array}{l} \ell_1 = 0 \\ \ell_2 = 0 \end{array} \right\}$$

$\ell_{1,2} : \mathbb{C}^4 \to \mathbb{C}$ — linear forms.



Original source details: GELFAND-P000742

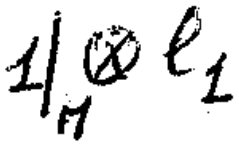
Diagram and typed description.

$$\begin{array}{ccc} I & \subset & H \otimes_{\mathbb{C}} \mathbb{C}^4 \\ \downarrow & & \square \downarrow 1|_H \otimes \ell_1 \\ I_{\ell_1} & \subset & H \end{array}$$



Original source details: GELFAND-P000742

Undeciphered passage. The heavily retraced mark to the left of the right-hand arrow is illegible.



Original source details: GELFAND-P000742

Uncertain reading. To the right of the arrow the label reads $1|_H \otimes \ell_1$.

Construct analogously I_{ℓ_2}.

Thus we have constructed 2 subspaces $I_{\ell_1}, I_{\ell_2} \subset H$

It turns out that

$$I_{\ell_1} + I_{\ell_2} \quad \text{[crossed out: depends?]}$$



Original source details: GELFAND-P000742

Uncertain reading. The crossed-out word to the right of the sum tentatively reads “зависит” (depends).

depends only on the σ-invar. plane $[z, \sigma(z)]$, i.e. a point of S^4

$$(I_{\ell_1} + I_{\ell_2})^{\perp} = \text{lin. subspace in } H, \text{ corresponding to } [z, \sigma(z)].$$

Condition 1. A bundle must result (the dimensions are maximal, i.e. the right ones), namely

$$\dim(I_{\ell_1} + I_{\ell_2})^{\perp} = r \tag{19}$$

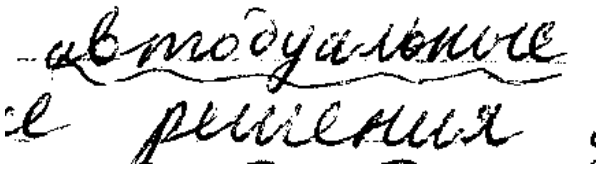
Condition 2. Hermitian property of the bundle.

The form $Q|_{I_{\ell_1}+I_{\ell_2}} > 0$

(a priori Q is not assumed to be > 0, but it turns out to do so under this condition).

A direct summand results. The connection is obtained by projecting the trivial one.

Theorem 1. This gives all [inserted above: self-dual] solutions of the Y.–M. equation, nonisom. bunds. correspond to inequivalent solutions.



Original source details: GELFAND-P000742

Uncertain reading. "Self-dual" is inserted above "solutions" and wavy-underlined.

One more condition has been omitted.

Take the eqs.:

$$\left\{ \begin{array}{l} \ell(z) = 0 \\ \ell(\sigma z) = 0 \end{array} \right\}$$

GELFAND-P000743

Condition 0.

$$Q(I_{\ell}, I_{\ell \circ \sigma}) = 0 \tag{20}$$

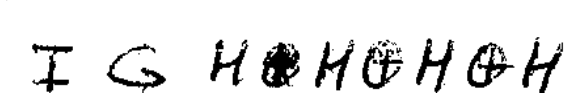
i.e. I_{ℓ} and $I_{\ell \circ \sigma}$ are orthogonal.

This gives a problem in linear algebra.

G. it is contained in an unpubl. work of Gelfand and Ponomarev on quadruples with an automorphism.

Ponomarev:

$$I \hookrightarrow H \oplus H \oplus H \oplus H \longrightarrow A \longrightarrow 0$$



Original source details: GELFAND-P000743

Uncertain reading. The first inclusion has an arrowhead. The first direct-sum sign is heavily retraced.

The images of the components H in A give 4 subspaces.

The complex invariants of the quadruple have been described.

Hermitian property: classify the quadruples invariant under the operation $\perp$



Uncertain reading. The abbreviation before "operation" tentatively reads "отн." (with respect to). The middle strokes are heavily retraced.

Original source details: GELFAND-P000743

a) Lagrangian subspaces, etc.



Diagram and typed description.

Original source details: GELFAND-P000743

M. Turning all this into numbers.

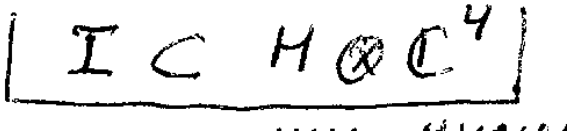


Diagram and typed description.

$$I \subset H \otimes \mathbb{C}^4$$

Original source details: GELFAND-P000743

— parameters of a multi-instanton solution

isomorphism of diagrams gives identical solutions.

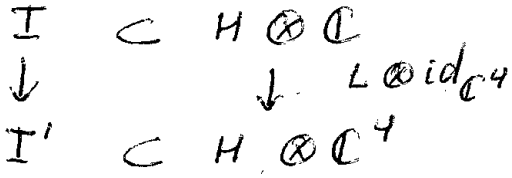


Diagram and typed description.

$$\begin{array}{ccc} I & \subset & H \otimes \mathbb{C} \\ \downarrow & & \downarrow L \otimes \mathrm{id}_{\mathbb{C}^4} \\ I' & \subset & H \otimes \mathbb{C}^4 \end{array}$$

Original source details: GELFAND-P000743



Uncertain reading. The upper row reads $H \otimes \mathbb{C}$, the lower row $H \otimes \mathbb{C}^4$.

Original source details: GELFAND-P000743

such isomorphisms of diagrams are allowed

L preserves the Hermitian form.

Scheme for computing $A_\mu(x)$.

$$(I \subset H \otimes \mathbb{C}^4, \quad e_1, \dots, e_r \text{ — a basis of } I) \longmapsto 4n \text{ vectors in } H$$
$$(e_{ij}), \quad i = 1, \dots, n; \quad j = 1, \dots, 4.$$
$$\dim H = 2n + r.$$



Uncertain reading. The last subscript in the list of basis vectors tentatively reads r.

Original source details: GELFAND-P000743

From them construct the matrix $\widetilde{Q}$

of $(n \times n)$ blocks of size 4×4

$(k\ell)$-th block at position (rs) has

$$\underset{\text{numerical matrix}}{Q(e_{kr}, e_{\ell s})}$$



Uncertain reading. The second index of the first argument is heavily retraced. The reading is tentatively e_{kr}.

Original source details: GELFAND-P000743

The matrix $\widetilde{Q}$ determines the whole situation

Assumption: $\operatorname{rk} \widetilde{Q} = 2n + r$ ($\Leftrightarrow L$ has no nontrivial horizontal sections) — we omit such degeneracies for simplicity.

GELFAND-P000744

[source number: 9]

Which $\widetilde{Q}$ are obtained:

a) $\widetilde{Q}$ is Hermitian;

b) $\operatorname{rk} \widetilde{Q} = 2n + r$;

c) “Condition 0”. This is a condition on all 4×4 blocks

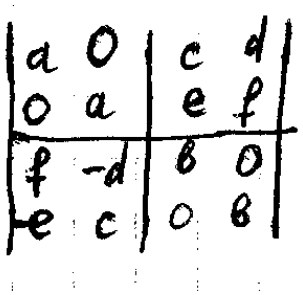


Diagram and typed description.

a	0	c	d
0	a	e	f
f	$-d$	b	0
$-e$	c	0	b

Original source details: GELFAND-P000744

scalar on the diag.;
symmetric with resp. to the diag.



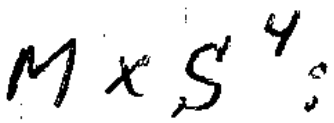
Uncertain reading. The crossed-out word tentatively reads “блоки” (blocks). Its middle is obscured by strokes.

Original source details: GELFAND-P000744

are related as written.
Conditions 1 and 2 are omitted (they too can be written out).
Next, the following are used: e_{ij}, but they do not actually enter the equations.

$$H \times S^4 = L \oplus L^{\perp}$$

$L^{\perp}$ is generated globally by $2n$ sections of the bundle $M \times S^4$;



Uncertain reading. This line reads $M \times S^4$.

Original source details: GELFAND-P000744

$$(21)_1 \qquad h_{k1} = (x_2 + ix_1)e_{k1} + (x_4 - ix_3)e_{k2} + e_{k3}$$
$$(21)_2 \qquad h_{k2} = -(x_4 + ix_3)e_{k1} + (x_2 - ix_1)e_{k2} + e_{k4}$$

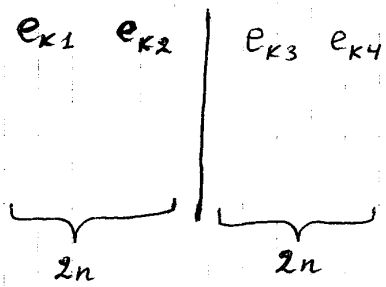


Diagram and typed description.

e_{k1} e_{k2} | e_{k3} e_{k4}

$2n$ $2n$

Original source details: GELFAND-P000744

At infinity everything becomes trivial.

“An instanton lives only a short while, and in imaginary time at that —
— a humanoid formation”

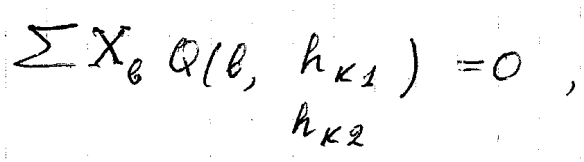
Choose from among e_{ij} a basis of H, $\quad B \subset (e_{ij})$

$\Leftrightarrow (2n+r)$-diagonal minor in $\widetilde{Q}$

(one such minor)

Using it, expand the system of generating sections of L.

$$(22) \qquad \sum_{\substack{h_{k1}\\ h_{k2}}} X_b\, Q(b, h_{k1}) = 0, \qquad k = 1, \dots, n, \qquad b \in B.$$



Uncertain reading. The manuscript places h_{k2} below h_{k1}, beneath the parentheses.

Original source details: GELFAND-P000744

this is a system of $2n$ eqs. for $2n+r$ unknowns X_b.

Fund. system of solutions $X^{(i)}$, $i = 1, \dots, r$.

Orthonormalize it to obtain $[Y^{(i)}]$ and now write down the answer:

GELFAND-P000745

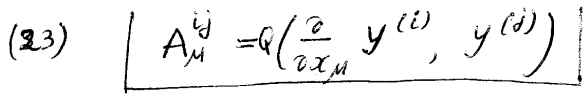
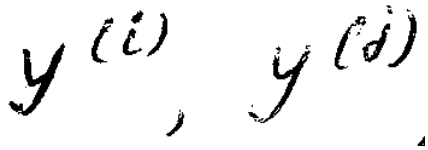


Diagram and typed description.

$$(23) \qquad \boxed{A_\mu^{ij} = Q\left(\frac{\partial}{\partial x_\mu} y^{(i)}, y^{(j)}\right)}$$

Original source details: GELFAND-P000745



Uncertain reading. Both letters in (23) have descenders and read as lowercase y.

Original source details: GELFAND-P000745

$X^{(i)}$ — rational functions with values in H;

roots of rational functions arise in orthonormalization.

Geometric theorem.

The diagrams $I \subset H \otimes \mathbb{C}^4$ with certain cond. ...

classify bundles over $\mathbb{C}P^3$ with conditions.

1) $H^i(L(-2)) = 0$ and likewise for L^*;



Uncertain reading. The dot of the superscript i is visible along the thin horizontal streak in the scan.

Original source details: GELFAND-P000745

2) $H^2(L(-1)) = 0$;

3) $H^0(L(-1)) = 0$.

GELFAND-P000746

[source number: 15]

28/XI-77. Gelfand Sem.

L. D. Faddeev. Solitons as particles.

I. $u(x,t)$ — a field, $x \in \mathbb{R}^d$, $d = 1, 2, 3$.

$$\Box = \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}$$

$$\Box u + m^2 u + \varepsilon V'(u) = 0 \tag{1}$$

$$\mathcal{L} = \int \left[\tfrac{1}{2}\,\partial u \cdot \partial u - \tfrac{1}{2} m^2 u^2 - \varepsilon V(u)\right] dx \tag{2}$$

$$\partial_0 u(x,t) \underset{\text{def}}{=} \pi(x,t)$$

$$\{u(x,t), \pi(y,t)\} = \delta(x-y) \tag{3}$$

$$H = \int \left[\tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\nabla u)^2 + \tfrac{1}{2} m^2 u^2 + \varepsilon V(u)\right] dx \tag{4}$$

Free field: $\varepsilon = 0$

In quantum theory u, π — operators.

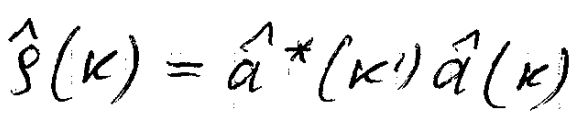
Fourier transf. for $\varepsilon = 0$:

$$H = \int \sqrt{k^2 + m^2}\, a^*(k) a(k)\, dk \tag{5}$$

$$\{a^*(k), a(k')\} = i\delta(k - k') \tag{6}$$

Quantization: let a, a^* — operators, the operator

$$\hat{\rho}(k) = \hat{a}^*(k')\hat{a}(k) \tag{7}$$

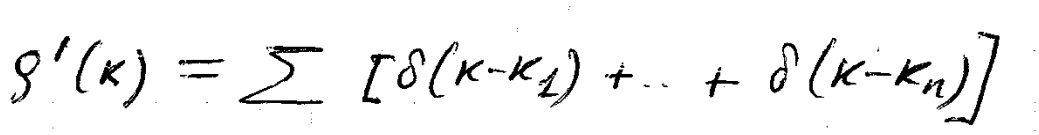


Uncertain reading. A prime is visible on k' in the first factor. The k in the second factor has no prime.

Original source details: GELFAND-P000746

has eigenvalues:

$$\rho'(k) = \sum \left[\delta(k - k_1) + \cdots + \delta(k - k_n)\right] \tag{8}$$



Original source details: GELFAND-P000746

Uncertain reading. After the equals sign comes $\sum$ without limits, then a square bracket.

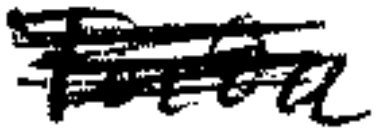
The spectrum of the Hamiltonian is interpreted in terms of particles.

$$E(k_1, \ldots, k_n) = \omega(k_1) + \cdots + \omega(k_n), \qquad \omega = \sqrt{k^2 + m^2} \tag{9}$$

(see: analogy with the oscillator)

GELFAND-P000747

In classical terms: if a finite-energy soln. of eq. (1) asymptotically approaches at ∞ a soln. of the free eq., then this picture is natural.



Uncertain reading. The crossed-out word tentatively reads “Тогда” (then).

Original source details: GELFAND-P000747

Then $H(u) = H(u_0)$, since H — an integral.

Solitons are a counterexample to this hypothesis.

Solitons: “everything that fails to satisfy the condition $u \to u_0$ as $|t| \to \infty$”

“Definition”: a soliton is something that



Uncertain reading. The crossed-out fragment tentatively reads “не уд” (beginning of “does not satisfy”).

Original source details: GELFAND-P000747

does not spread out at infinity while having finite energy.

Example: a solution of the stationary eq..

$u_s(x)$ — one solution $\Rightarrow$ a Lorentz transf. gives



Uncertain reading. A heavily crossed-out entry precedes the left parenthesis. u_s is written above it.

Original source details: GELFAND-P000747

$$u_s\left(\frac{x - x_0 - vt}{\sqrt{1 - v^2}}\right) = u_s(x, t \mid x_0, v) \tag{10}$$



Uncertain reading. There is a small crossed-out mark at the right end of the fraction bar. The parenthesis is retraced and extended downward.

Original source details: GELFAND-P000747

already a soliton, depending on 2 parameters.

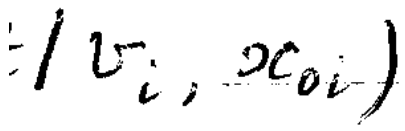
At large times: if $v_1 \neq v_2$ then

$$u(x,t \mid x_{01}, v_1) + u(x,t \mid x_{02}, v_2) \tag{11}$$

is asymptotically a solution.

Then the asymptotics: as $|t| \to \infty$

$$u \to u_0 + \sum u(x,t \mid v_i, x_{0i}) \tag{12}$$



Uncertain reading. After the vertical bar the manuscript reads v_i, x_{0i}.

Original source details: GELFAND-P000747

(a hypothesis in general; a theorem in models)

As $t \to +\infty$ and as $t \to -\infty$ they differ.

Energy

$$H(u) = H(u_0) + \sum_{i=1}^{n} H_s(v_i) \tag{13}$$

It is clear from the relativistic formula that

$$H_s(v) = \frac{M}{\sqrt{1-v^2}} = \sqrt{p^2 + M^2} \qquad \text{(“soliton energy”)} \tag{14}$$

M — the energy of a static soliton (“mass” by definition).

(14) — the definition of p.

This gives 2 parts of phase space. Energy:

$$\int \omega(k)\rho(k)\,dk + \sum_{i}^{n} \sqrt{p_i^2 + M^2} \tag{15}$$

Solitons contributed already in the classical theory. In quantum mechanics there enters $\frac{M}{\hbar c}$; solitons persist as $\hbar \to +0$.

GELFAND-P000748

[source number: 16]

In quantum mechanics the spectrum will be

$$\sum_{i=1}^{N} \sqrt{k_i^2 + m^2} + \sum_{i}^{n} \sqrt{p_i^2 + M^2} \tag{16}$$

From one field, 2 kinds of particles (semiclassically)

Periodic solution.

$$u_p(x, t \mid x_0, \bar{v}, \beta, \alpha) \tag{17}$$



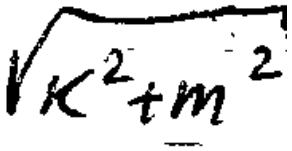
Uncertain reading. The retraced sign tentatively reads β.

Original source details: GELFAND-P000748

β is related to T (period), α (initial phase) are parameters

In this case

$$\sum_{i=1}^{N} \sqrt{k^2 + m^2} + \sum_{i}^{n} \sqrt{p_i^2 + M^2} + \sum_{k}^{\tau} \sqrt{p_k^2 + M(\alpha_k)} \tag{18}$$



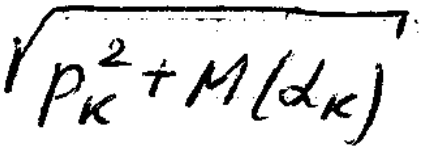
Uncertain reading. The first radicand reads k^2, with no visible subscript.

Original source details: GELFAND-P000748



Uncertain reading. The last upper summation limit tentatively reads τ. Another possible reading is r.

Original source details: GELFAND-P000748



Uncertain reading. The last radicand has $M(\alpha_k)$, with no visible exponent 2.

Original source details: GELFAND-P000748

The main model problem:

$$v(u) = 1 - \cos u - \frac{u^2}{2} \tag{19}$$



Uncertain reading. The initial sign before (u) is rendered as v. It may be υ.

Original source details: GELFAND-P000748

The eq. takes the form:

$$\Box u + m^2 \sin u = 0 \tag{20}$$

(the $(\sin\varphi)_2$)

One more parameter: in the Poisson bracket put

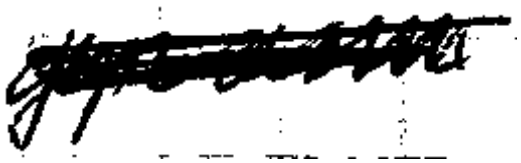
$$\{\partial_0 u, u\} = \gamma\delta(\ldots) \tag{21}$$

Then it turns out that

$$M = \frac{8m}{\gamma} \qquad \text{(“heavy particle”)} \tag{22}$$

$$M_2(\alpha) = \frac{16m}{\gamma}\sin\alpha, \qquad 0 < \alpha \le \frac{\pi}{2}. \tag{23}$$

These



Original source details: GELFAND-P000748

Uncertain reading. The crossed-out word tentatively reads “двумя” (two).

solutions determine everything.

Here one must take

$$\alpha = \frac{\gamma}{16}n, \quad n = 1, 2, \ldots \tag{24}$$

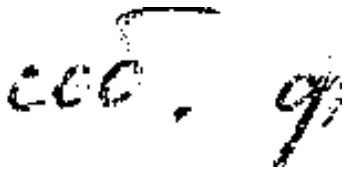
(Takhtajan, Faddeev, ≈ 1974)

A detailed review (in English) should appear shortly.

GELFAND-P000749

II. Consistent quantization and the S-matrix.

It has not yet been possible to find the actual eigenfunctions.



Original source details: GELFAND-P000749

Uncertain reading. The abbreviation in the first sentence reads “соб.” (eigenfunctions).

But the S-matrix has been computed! (by an indirect route: the path integral)
Definitions: write

$$H = \int \left[\frac{\pi^2}{2} + \frac{(\nabla u)^2}{2} + \frac{m^2}{2} u^2 + V(u) \right] \tag{25}$$

and take π and u to be operators with the relation

$$[\hat{\pi}(x), \hat{u}(y)] = i\delta(x - y) \tag{26}$$

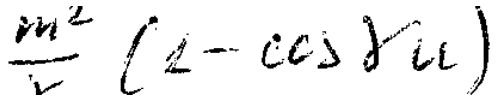
There is no contradiction, but one must begin with renormalizations.
Glimm, Jaffe:
Frohlich



Original source details: GELFAND-P000749

Uncertain reading. The middle letters of the surname are unclear. The tentative reading is "Frohlich".

$$\begin{bmatrix} \frac{m^2}{\gamma}(1 - \cos \gamma u) & \text{for small } \gamma & \text{constructed} \\ \text{the Hamiltonian rigorously} & (\text{for } x \in \mathbb{R}^1). \end{bmatrix}$$



Original source details: GELFAND-P000749

Uncertain reading. The denominator and the factor before u read γ, with no visible exponent 2.

Lemma 1 (from ordinary quantum mechanics). Consider the eq.

$$i\frac{\partial \Psi}{\partial t} = H\Psi \tag{27}$$

$$H = -\frac{\Delta}{2m} + v(x), \quad v \text{ decays rapidly} \tag{28}$$



Original source details: GELFAND-P000749

Uncertain reading. The rounded sign before (x) and later in the line is rendered as v. It may be v.

Then the limit exists

$$S = \lim_{\substack{t'' \to \infty \\ t' \to -\infty}} e^{iH_0 t''} e^{-iH(t''-t')} e^{-iH_0 t'} \tag{29}$$

(S-matrix) and S — a unitary operator.
It is better to work in the momentum representation: the kernel is denoted by $S(p'', p')$. It suffices to know the kernel of the evolution op. $e^{-iH(t''-t')} = K$

$$K(x'', t'' \mid x', t')$$

One must take

$$\begin{cases} x'' = \frac{p''}{m} t'' \\ x' = \frac{p'}{m} t' \end{cases} \tag{30}$$

(by the classical formulas with the free H_0). Then

$$S(p'',p') = \lim_{\substack{t''\to+\infty \\ t'\to-\infty}} \frac{K\left(\frac{p''}{m}t'',t'' \mid \frac{p'}{m}t',t'\right)}{K_0\left(\frac{p''}{m}t'',t'' \mid 0,0\right) K_0\left(0,0 \mid \frac{p'}{m}t',t'\right)} \tag{31}$$

(analogous to (29))

GELFAND-P000750

[source number: 17]

The right-hand side of (31) is computed by a path integral over trajectories with classical asymptotics.

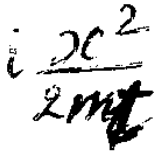
Comment on (31): apply it to Ψ, considering a wave packet

$$\Psi(x,t) = \left[\frac{1}{2\pi}\int \Psi(p) e^{ipx} e^{-i\frac{p^2}{2m}t}\, dp\right] \underset{\substack{\uparrow \\ \text{stationary-phase} \\ \text{method}}}{\cong} \widetilde{\Psi}\left(\frac{mx}{t}\right) e^{i\frac{x^2}{2mt}} \tag{32}$$



Uncertain reading. A short mark appears above the Ψ under the integral. A tilde is uncertain.

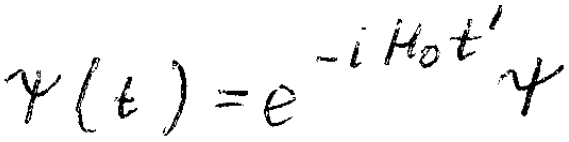
Original source details: GELFAND-P000750



Uncertain reading. The last denominator symbol in the exponential is heavily retraced and tentatively reads t.

Original source details: GELFAND-P000750

$$\Psi(t) = e^{-iH_0t'}\Psi$$



Uncertain reading. The left side reads $\Psi(t)$, while the exponential on the right has t'.

Original source details: GELFAND-P000750

We obtain

$$K(\ldots \mid \tfrac{p}{m}t,t)\,\widetilde{\Psi}(p)\sqrt{\frac{m}{t}}\,dp \tag{33}$$

with factors (in each part we pass to its own coordinate system).

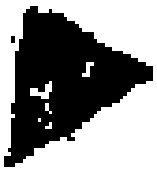
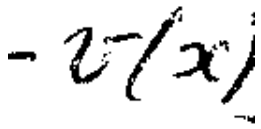


Diagram and typed description. ▶

Original source details: GELFAND-P000750

Path integral. First for quantum mechanics

$$K(x'', t'' \mid x', t') = \frac{1}{N} \int e^{i \int_{t'}^{t''} \left[\frac{m\dot{x}^2}{2} - v(x)\right] dt} \prod_{t' \le t \le t''} dx(t), \tag{34}$$



Uncertain reading. The first sign in $v(x)$ is rounded. It may be v.

Original source details: GELFAND-P000750

over trajectories such that

$$x(t') = x', \quad x(t'') = x'' \tag{35}$$

N — a normalization factor (depending on time).

Now one must take $x(t') = \frac{pt'}{m}$, $x(t'') = \frac{pt''}{m}$.

In field theory one must write

$$K(u''(x), t'' \mid u'(x), t') = \frac{1}{N} \int \exp\left\{ i \int \mathcal{L}(x)\, dx \right\} \prod_{\substack{t' \le t \le t'' \\ -\infty < x < \infty}} du\, [\text{crossed out}](x, t) \tag{36}$$



Undeciphered passage. There is a crossed-out fragment after du and before (x, t). Its reading is undetermined.

Original source details: GELFAND-P000750

over u, such that

$$\begin{cases} u(x, t'') = u''(x) \\ u(x, t') = u'(x) \end{cases} \tag{37}$$

For the initial and final configurations one must take sums of solitons. This gives transition probabilities.

There is as yet no good quantum version of the inverse problem.



Uncertain reading. A thin diagonal stroke crosses the end of “Хорошего” (good) and continues downward to the right.

Original source details: GELFAND-P000750

For the S-matrix of sine-Gordon, an exact formula has been guessed (Zamolodchikov from ITEP; see Comm. on Math. Physics)



Undeciphered passage. Crossed-out strokes and part of a rectangular box precede “Гипотеза” (hypothesis). Their reading is undetermined.

Original source details: GELFAND-P000750

Hypothesis: all classically solvable problems are also solvable in the quantum version

$$\langle u''(x) \mid e^{-iH(t''-t')} \mid u'(x)\rangle \tag{38}$$

— a matrix element given by the path integral. Now one must divide and let t tend to $\pm\infty$. Corrections to the



Undeciphered passage. A short fragment between “к” (to) and “классической” (classical) is crossed out and undeciphered.

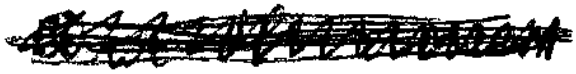
Original source details: GELFAND-P000751

classical mass arise already for a single soliton.

Problems arising in the computation of the integral.

The stationary-phase method.

$$u(x,t) = U_{cl}(x,t) + \sqrt{h}\,\varphi(x,t) + \dots \tag{39}$$



Undeciphered passage. A phrase preceding “Возникает оператор” (the operator arises) is heavily crossed out. Its reading is undetermined.

Original source details: GELFAND-P000751

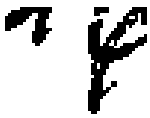
The operator arises

$$A\varphi = \Box\varphi + m^2\varphi + V''(u_\varphi)\varphi \tag{40}$$



Uncertain reading. The heavily retraced signs after m^2 tentatively read $\varphi+$.

Original source details: GELFAND-P000751



Uncertain reading. The subscript on u tentatively reads φ.

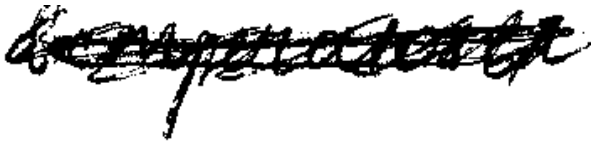
Original source details: GELFAND-P000751

Corrections are computed through $\det A$.

It is expressed through the S-matrix for an auxiliary linear equation. Loops arise next. One needs the inverse operator $R = A^{-1}$ and the kernel $R(x,y)$, namely

$$\int R^3(x,x)\,d^2x \tag{41}$$

Hopes: it is possible that such functionals can be computed explicitly.



Uncertain reading. The crossed-out phrase tentatively reads “в квадратурах” (by quadratures).

Original source details: GELFAND-P000751

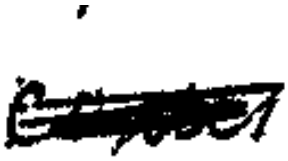
Let u_1 — a one-soliton solution (stationary),

$$V''(u_1(x)) = q(x). \tag{42}$$

Computing

$$\det\left(\frac{d^2}{dt^2} - \frac{d^2}{dx^2} + q(x)\right) \tag{43}$$

reduces to the spectral density of the stationary op. (the variables separate). One obtains $q(x) = \frac{1}{\mathrm{ch}^2\, x}$ and all eigenfunctions are found explicitly (the simplest Bargmann potential).
$\det A$ is expr. through the eigenfunction with zero eigenvalue.
Corrections to the mass of the



Original source details: GELFAND-P000751

Undeciphered passage. A short word before “солитона” (soliton) is heavily crossed out and undeciphered.

soliton:

$$M = \frac{8m}{\gamma} - \frac{m}{\pi} + O(\gamma) \tag{44}$$

Conjecture: for sine-Gordon

$$M = \frac{8m}{\gamma} - \frac{m}{\pi}, \tag{45}$$

i.e. there are no higher corrections. (justif. by Zamolodchikov)



Original source details: GELFAND-P000751

Uncertain reading. The abbreviation before the surname tentatively reads “оправд.” (justified). Its first letter may be “с”.

For $\gamma \leq 8\pi$ everything is fine, and beyond this the soliton disappears.

GELFAND-P000752

[source number: 18]

Formulas for the S-matrix of sine-Gordon.
One must compute the abbreviated action for the 2-soliton solution.

$$S_{sa}(p_1'', p_2''; p_1', p_2') = \delta(p_1'' - p_1')\delta(p_2'' - p_2')e^{i\Theta(p_1, p_2)} \tag{46}$$



Original source details: GELFAND-P000752

Uncertain reading. The separator between the arguments is heavily retraced. It is tentatively rendered as a semicolon.

(There is a soliton and an antisoliton)

The action (classical)

$$\Theta = \frac{8}{\gamma}\int_0^{\pi} d\alpha \left[\ln \frac{e^{i(\beta-\alpha)}+1}{e^{i\beta}+e^{-i\alpha}}\right] \tag{47}$$

[Crossed out: For $\gamma = \frac{8\pi}{N}$, N — an integer]



Original source details: GELFAND-P000752

Undeciphered passage. The end of the crossed-out line following “целое” (integer) is undeciphered.

(the integral cannot be evaluated).

One must take $\gamma = \frac{8\pi}{N}$, N — an integer

$$\Theta \longrightarrow \sum \ln(\dots) \tag{48}$$

Zamolodchikov took this to hold for $\gamma = \frac{8\pi}{N}$ and interpolated.

As a result sine-Gordon turned out to resemble a quantum-mechanical system [Crossed out: and] with

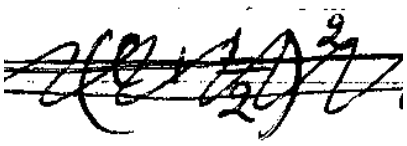
$$H = -\frac{d^2}{dx^2} + \frac{a}{\mathrm{ch}^2\,\alpha x} \tag{49}$$

Uncertain reading. The numerator tentatively reads a, possibly q.



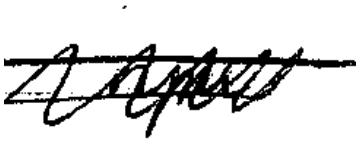
Original source details: GELFAND-P000752

[Crossed out: Eigenvalue $\alpha(\ell+\frac{1}{2})^2$]



Uncertain reading. The crossed-out formula tentatively reads $\alpha(\ell+\frac{1}{2})^2$.

Original source details: GELFAND-P000752



Undeciphered passage. The end of the crossed-out line is undeciphered. A fragment tentatively reads “с пр...”.

Original source details: GELFAND-P000752

A remarkable dynamical system

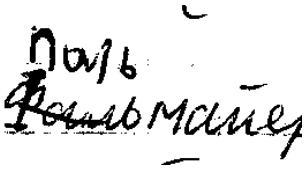
is associated with a field (chiral)

$$n(x) : R^2 \longrightarrow S^n \subset R^{n+1} \tag{50}$$

$$\mathcal{L} = \frac{1}{2} \int \partial n \cdot \partial n \, dx \tag{51}$$

it is nontrivial because of its values (there is nonlinearity).

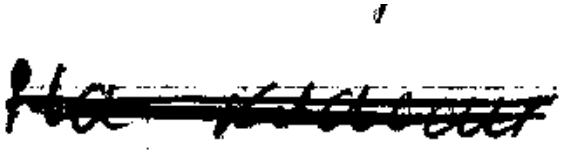
L-A pair has been written down ([Crossed out: Fol; written above: Pol]meyer (?); Zakharov and Mikhailov)



Original source details: GELFAND-P000752

Uncertain reading. The beginning of the surname is corrected from tentative “Фоль” (Fol) to “Поль” (Pol).

[Crossed out: Na]



Original source details: GELFAND-P000752

Undeciphered passage. The remainder of the crossed-out phrase before “Квантование” (quantization) is undeciphered.

Quantization is weakly connected with the classical theory.

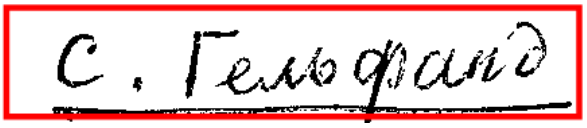
Massive particles appear as solitons.

There will be a talk on this at the seminar named after I. G. Petrovsky.

GELFAND-P000753

13/II-78. Gelfand Seminar.

I. M. G. there are 2 primary concepts in category theory: 1) Topos 2) Derived category.



Original source details: GELFAND-P000753

Diagram and typed description. S. Gelfand

Problem: describe all algebraic bundles over



Original source details: GELFAND-P000753

Undeciphered passage. A short sign before the projective-space notation is heavily crossed out.

$\mathbb{P}^n$. (n-dimensional projective space; for example, over $\mathbb{C}$).

Ξ — affine space.

$$\xi \sim \alpha\xi, \quad \alpha \in \mathbb{C}^*, \quad (\xi \neq 0); \qquad \text{Then } \Xi/\sim = \mathbb{P}^n.$$



Original source details: GELFAND-P000753

Uncertain reading. The multiplier and the following sign are tentatively read as α.

Λ — the Grassmann algebra in $(n+1)$-th variable (



Original source details: GELFAND-P000753

Undeciphered passage. An abbreviation at the beginning of the parentheses is heavily crossed out; its reading is unresolved.

algebra over Ξ).

$$\Lambda = \{\xi_0, \dots, \xi_n \mid \xi_i\xi_j + \xi_j\xi_i = 0\}.$$

$$\dim \Lambda = 2^{n+1}.$$

Introduce a grading on Λ:

$$\Lambda = \Lambda_0 \oplus \Lambda_{-1} \oplus \cdots \oplus \Lambda_{-(n+1)}, \qquad \text{[Crossed out: } \Lambda_0 \text{ — the field of constants]} \tag{1}$$

where

$$\Lambda_{-k} \text{ [Crossed out: } \Lambda_k?] = \{\xi_{i_1} \cdots \xi_{i_k}\}, \qquad \Lambda_0 \text{ — the field of constants.}$$



Original source details: GELFAND-P000753

Uncertain reading. The cancelled expression before the equality sign tentatively reads Λ_k.

Objects: a graded finite-dimensional representation of Λ or a graded Λ-module:

$$V = \bigoplus V_j, \qquad \dim V < \infty. \tag{2}$$

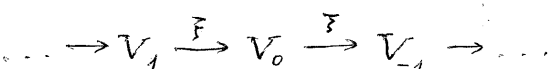
$$\Lambda_{-k} V_j \subset V_{j-k}. \tag{3}$$

Let $\xi_0, \dots, \xi_n$ — be a basis in Ξ, $\xi \in \Xi$.

$$\xi = c_0\xi_0 + \cdots + c_n\xi_n. \tag{4}$$

$(-k)$ — for historical reasons (from Serre's paper).

Then $\xi : V_j \longrightarrow V_{j-1}$. Hence, since $\deg \xi = -1$, we have



Original source details: GELFAND-P000753

Diagram and typed description.

$$\cdots \longrightarrow V_1 \xrightarrow{\xi} V_0 \xrightarrow{\xi} V_{-1} \longrightarrow \cdots$$

$\xi^2 = 0 \Rightarrow$ we have obtained a complex.
Thus, from an element $\xi \neq 0$ and a module V one constructs the complex

$$L_\xi(V) = \{\cdots \longrightarrow V_1 \xrightarrow{\xi} V_0 \xrightarrow{\xi} V_{-1} \longrightarrow \cdots\}. \tag{5}$$

Definition. V is called proper, if

$$\forall \xi \neq 0, \qquad H_i(L_\xi(V)) = 0 \quad \text{for } i \neq 0. \tag{6}$$

Here:

$$H_i \text{ [Crossed out: ?] } (L_\xi(V)) = \ker(V_i \longrightarrow V_{i-1}) / \operatorname{Im}(V_{i+1} \longrightarrow V_i) \tag{7}$$



Original source details: GELFAND-P000753

Undeciphered passage. A sign between the index and the parenthesis in (7) is heavily crossed out.



Original source details: GELFAND-P000753

Uncertain reading. The equation number is heavily retraced; it tentatively reads (7).

— [Crossed out: quotient spaces?]



Original source details: GELFAND-P000753

Uncertain reading. The crossed-out word tentatively reads “факторпространства” (quotient spaces); its beginning is obscured by dense strokes.

homology of the complex.

GELFAND-P000754

[source number: 43]

In this case

$$\dim H_0^?(L_\xi(V)) = \text{const}$$



Original source details: GELFAND-P000754

Uncertain reading. A dense mark appears above H_0; its internal strokes cannot be distinguished. A question mark represents it in the typeset formula.

(does not depend on ξ)

by Euler's formula

(left side — the Euler characteristic ξ?



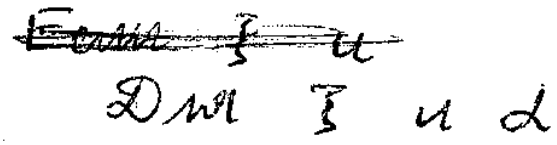
Original source details: GELFAND-P000754

Uncertain reading. The overwritten mark after "характеристика" (characteristic) may be ξ.

equal to $\sum(-1)^i \dim V_i$.)

Bundle: $\xi \longmapsto H_0(L_\xi(V))$.

[Crossed out: If ξ and]



Original source details: GELFAND-P000754

Uncertain reading. The short crossed-out line tentatively reads "Если ξ и" (If ξ and).

For ξ and $\alpha\xi$ it is the same (there is a natural isomorphism).

Example. $V = \Lambda$ (the regular representation).

$V_j = \Lambda_j$ (the same grading as before).

Answer: $H_i(L_\xi(V)) = 0, \quad \forall i, \quad \forall \xi \neq 0$.

We obtain the zero bundle.

Direct sums give the same result.

$V_j = \Lambda_{j+\ell} \quad (\ell$ — fixed$)$

$\Rightarrow$ we again obtain a trivial bundle.

Hence free modules lead to trivial bundles.

Theorem 1 Every bundle is obtained by such a construction, and V is determined uniquely up to adding a free module.

(Bernstein, S. Gelfand, I. Gelfand; independently a similar result by Beilinson, a 3rd-year student)

Refinement of the bundle construction.

V — an arbitrary Λ-module.

$O(k)$ — the standard bundles (their sections are homogeneous rational functions of degree $k \in \mathbb{Z}$), one-dimensional.

Consider

$$\mathcal{L}_j = V_{-j} \otimes O(j). \tag{8}$$

— a bundle over $\mathbb{P}^n$; each fiber is multiplied by V_{-j}.

$O(1)$ — the one that has a section.

Construct

$$d : \mathcal{L}_j \longrightarrow \mathcal{L}_{j+1}. \tag{9}$$

$f(\xi)$ — a section of $\mathcal{L}_j$ over an open set

a homogeneous function of degree j with values in V_{-j}

$$d(f(\xi)) = \xi(f(\xi)). \tag{10}$$

(on the right is the action of ξ on V_{-j}, giving V_{-j-1}).

GELFAND-P000755

$O(1)$ — the Hopf bundle. (or $O(-1)$). We obtain the complex

$$\{\ldots \longrightarrow \mathcal{L}_{-1} \xrightarrow{d} \mathcal{L}_0 \xrightarrow{d} \mathcal{L}_1 \longrightarrow \ldots\} = \mathcal{L}(V)\,\xi?. \tag{11}$$



Uncertain reading. The overwritten mark after $\mathcal{L}(V)$ tentatively reads ξ; it may be a cancellation.

Original source details: GELFAND-P000755

over each point ξ there is the same complex as before (easy to check)

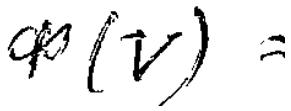
Assertion: if V — proper, then $H^0(\mathcal{L}(V))$ — a bundle and $H^i(\mathcal{L}(V)) = 0$ for $i \neq 0$.

Set

$$\Phi(V) = H^0(\mathcal{L}(V)). \tag{12}$$

Theorem $1'$. 1) Every bundle is $\Phi(V)$ for some proper V

2) $\varphi(V) \approx \Phi(V') \Leftrightarrow \exists$ free P, P': $V \oplus P \cong V' \oplus P'$.



Uncertain reading. The left side of part 2 has φ, while the right side has Φ. Both letters in (15) have the form φ.

Original source details: GELFAND-P000755

The inverse map.

From a bundle one must construct a Λ-module V.

$$\underset{\text{bundle}}{\mathcal{E}} \rightsquigarrow \mathcal{E} \otimes O(1) \tag{13}$$

what does this correspond to in Λ-modules?

Suppose there is an exact sequence of Λ-modules

$$0 \longrightarrow \underset{\text{proper}}{V} \longrightarrow \underset{\text{free}}{P} \longrightarrow V' \longrightarrow 0 \tag{14}$$

Construct $W \circ ?$; $W_j = V'_{j+1}$.



Uncertain reading. The small separate mark after W resembles $\circ$ crossed by a horizontal stroke. Its function is unresolved.

Original source details: GELFAND-P000755

$$\underline{\text{Assert.}}: \quad W \text{ — proper and } \varphi(W) = \varphi(V) \otimes O(1) \tag{15}$$

Let $\mathcal{E}$ — a bundle; consider $\mathcal{E} \otimes O(N)$ for sufficiently large N. Serre's theorem: a bundle $\mathcal{E} \otimes O(N)$ has many sections ("an ample bundle"), they pass through any point in any fiber.

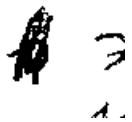
Consider $C_N = \Gamma(\mathbb{P}^n, \mathcal{E} \otimes O(N))$;

$$\dim \Gamma(\mathbb{P}^n, O(1)) = n + 1 \tag{16}$$

(linear functions of ξ)

$\xi_0, \ldots, \xi_n$ — a basis in Ξ

$x_0, \ldots, x_n$ — the dual basis in $\Gamma(\mathbb{P}^n, O(1))$ ([crossed out: v?]



Original source details: GELFAND-P000755

Uncertain reading. A heavily cancelled mark before "это пр-во" (this is the space) may be the letter "в".

this is the space of linear functions)

(coefficients of the expansion in the preceding basis)

There are operators

$$X_i : C_N \longrightarrow C_{N+1}, \quad i = 0, 1, \ldots, n. \tag{17}$$

(multiplication by x_i)

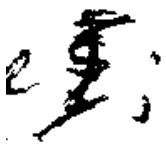
GELFAND-P000756

[source number: 44]

Now construct 2 Λ-modules

$$V = C_N \otimes \Lambda \tag{18}$$

(on C_N the action is trivial Ξ?;



Original source details: GELFAND-P000756

Uncertain reading. The sign following "действие" (action) is heavily retraced; the reading Ξ is tentative.

on Λ the usual action)

$$V' = C_{N+1} \otimes \Lambda. \tag{18'}$$

We have

$$\partial : V \longrightarrow V' \tag{19}$$

$$\partial(c \otimes \lambda) = \sum_{i=0}^{n} X_i(c) \otimes \xi_i \lambda \tag{20}$$

The map ∂ does not depend on the basis. Set $W = \operatorname{Ker} \partial$.

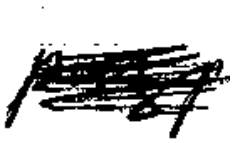
This is a Λ-module. Now one must shift back N times

$$0 \longrightarrow W' \longrightarrow P \longrightarrow W \longrightarrow 0$$

$W \longmapsto W'$ perform N times (with a shift of grading).

This is precisely that representation (that same Λ-module)

I.M.G. Classification of bundles — not quite the right problem (they do not form an abelian category). One must allow the fibers to change



Undeciphered passage. A short word between “менять” (change) and “размерность” (dimension) is heavily crossed out.

Original source details: GELFAND-P000756

dimension — introduce sheaves. The work being presented develops Serre’s theory.

Serre’s theorem

$$\{\text{Sheaves on } \mathbb{C}P^n\} \Leftrightarrow \left\{ \begin{array}{l} \text{finitely generated } \overset{\text{(graded)}}{\text{modules}} \\ \text{over } \mathbb{C}[x_0, \ldots, x_n] \end{array} \Big/ \begin{array}{l} \text{up to} \\ \text{a finite} \\ \text{number of homogeneous} \\ \text{components} \end{array} \right\}$$

ξ?



Uncertain reading. The separate overwritten sign to the left of the reference may be ξ; its function is unresolved.

Original source details: GELFAND-P000756

(an article in the collection “Fiber spaces”; 1953).



Undeciphered passage. A short fragment after the collection reference is crossed out.

Original source details: GELFAND-P000756

Construct a sheaf from the module M.

Define its sections over $E_i = \{x.\ x_i \neq 0\}$

[Crossed out: $\Gamma(E_i, M)$?]



Uncertain reading. The cancelled formula tentatively reads $\Gamma(E_i, M)$; diagonal strokes obscure the arguments.

Original source details: GELFAND-P000756

$$\Gamma(E_i, \underset{\text{our sheaf}}{L}) = \left\{ \begin{matrix} \dfrac{m}{x_i^s}, & \deg m = s \\ & m \in M \end{matrix} \right\}.$$

$O(N)$ arises from $M = \mathbb{C}[x_0, \ldots, x_n]$ with shifted grading

$$M_i = \{\mathbb{C}[x_0, \ldots, x_n]\}_{i+N}.$$

GELFAND-P000757

27/II-78. Gelfand Seminar.

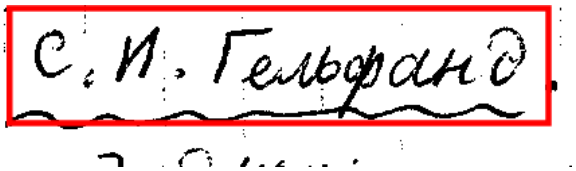


Diagram and typed description. S. I. Gelfand

Original source details: GELFAND-P000757

Problem: describe algebraic vector bundles over projective space.

Ξ — $(n+1)$-dimensional space (vector).

$\mathbb{P} = \mathbb{P}(\Xi)$ — projective n-dimensional space

[Crossed out]



Undeciphered passage. A short word before “Выберем базис” (choose a basis) is crossed out; a complete reading is unresolved.

Original source details: GELFAND-P000757

Choose a basis of Ξ: $\xi_0, \ldots, \xi_n$

and let

$\Lambda = \Lambda(\Xi)$ — the exterior algebra of Ξ

$$\Lambda = \bigoplus_{i=-n-1}^{0} \Lambda_i$$

Λ has generators $\xi_0, \ldots, \xi_n$ and relations $\xi_i\xi_j + \xi_j\xi_i = 0, \quad \xi_i^2 = 0;$

$$\Lambda_{-k} = \{\xi_{i_1} \ldots \xi_{i_k}\}.$$

$$\dim \Lambda_{-k} = \binom{n+1}{k}$$

Remark: let $\xi \in \Lambda_{-1}$. Then $\xi^2 = 0$. (and this is equivalent to all the relations).

Object: a graded finite-dimensional Λ-module

$$V = \bigoplus_j V_j, \qquad \lambda \in \Lambda_i,\ v \in V_j \Rightarrow \lambda v \in V_{i+j}$$

(j may be arbitrary integers).

From each V construct a collection of vector bundles over $\mathbb{P}(\Xi)$ and their maps (a complex of bundles).

$\forall j$ construct the bundles $O(j)$

3 definitions of $O(j)$

1) transition functions $(z_k/z_\ell)^j$;

2) sections over an open set $=$ homogeneous rational functions of degree j;

3) j-th tensor power of the anti-Hopf bundle.

The corresponding sheaf can be defined as follows:

$$\Gamma(O(j), U) = \{\text{rational functions of degree } j \text{ without singularities in } \bar{U}\}$$

$U \subset \mathbb{P}^n$, $\bar{U}$ — the preimage of U in Ξ

Problem: why does a one-dimensional bundle result?

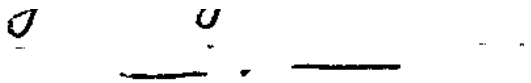
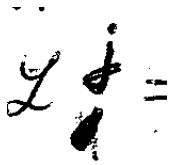


Diagram and typed description. —— · ——

Original source details: GELFAND-P000758

Set

$$\mathcal{L}^j = V_{-j} \otimes O(j) \qquad \text{— a bundle.}$$

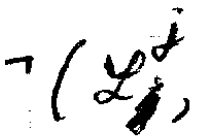


Uncertain reading. The subscript of $\mathcal{L}$ is blotted out; j is written above. The cancelled subscript tentatively reads j.

Original source details: GELFAND-P000758

V_{-j} means the constant bundle with fiber V_{-j}

$$\Gamma(\mathcal{L}^j, U) = \left\{ \begin{matrix} \text{rational functions with values in } V_{-j}, \\ \text{homogeneous degree } j, \text{ nonsingular in } \bar{U} \end{matrix} \right\}.$$



Uncertain reading. In $\Gamma(\mathcal{L}^j, U)$ the subscript is also blotted out; j is written above. The cancelled subscript tentatively reads j.

Original source details: GELFAND-P000758

Construct the maps:

$$d : \mathcal{L}^j \longrightarrow \mathcal{L}^{j+1}$$

On sections: let $f(\xi) \in \Gamma(\mathcal{L}^j, U)$,

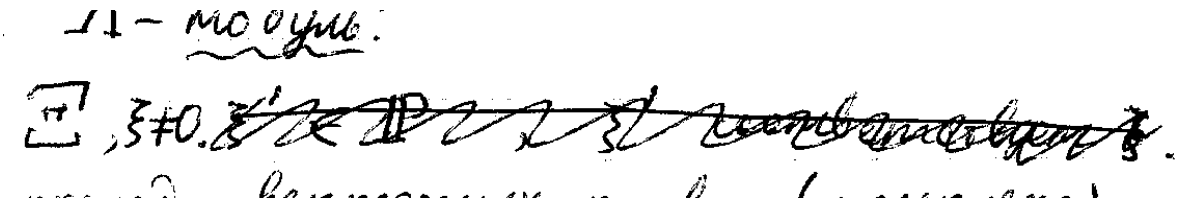
$$\xi \in \Lambda_{-1} = \Xi$$

$$(df)(\xi) = \xi[f(\xi)]$$

(on the right is the action of ξ as an element of the exterior algebra on V_{-j}).
A complex of bundles results:

$$\mathcal{L}(V) = \{\mathcal{L}^j, d\}, \qquad d^2 = 0.$$

Proper Λ-module:
let $\xi \in \Xi, \quad \xi \neq 0$. [Crossed out: $\xi' \in \mathbb{P}$ — ξ' corresponding to ξ.]



Original source details: GELFAND-P000758

Uncertain reading. The cancelled projective-point statement tentatively reads $\xi' \in \mathbb{P}$ — ξ' corresponding to ξ. Some crossed-out signs remain unclear.

Consider the sequence of vector spaces (complex)

$$\{\ldots \longrightarrow V_{-j+1} \xrightarrow{\xi} V_{-j} \xrightarrow{\xi} V_{-j-1} \longrightarrow \ldots\} = L_\xi.$$

GELFAND-P000759

[source number: 56]

Definition. V proper, if

$$H^i(L_\xi) = 0, \quad \forall i \neq 0, \quad \forall \xi \neq 0.$$

L_ξ — the complex obtained from $\mathcal{L}(V)$ at the point ξ.
For proper V

$$\dim H^0(L_\xi) = \text{const}$$

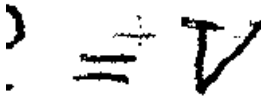
The spaces $H^0(L_\xi)$ glue into a vector bundle. The precise definition is as follows: set

$$\begin{aligned}\Phi(V) &= H^0(\mathcal{L}(V)) \\ &= \operatorname{Ker}(d : \mathcal{L}^0 \to \mathcal{L}^1) / \operatorname{Im}(d : \mathcal{L}^{-1} \to \mathcal{L}^0).\end{aligned}$$

Theorem 1. 1) Every algebraic vector bundle $\mathcal{E}$ has the form $\Phi(V)$ for some proper V.
2)

$$\Phi(V) \cong \Phi(V') \Longleftrightarrow V \oplus P \cong V' \oplus P'$$

for free P, P'.



Uncertain reading. In the second relation there is a faint stroke above the two horizontal lines: tentatively $\cong$, possibly $=$.

Original source details: GELFAND-P000759

The inverse map.

Coherent sheaves of modules on $\mathbb{P}^n$

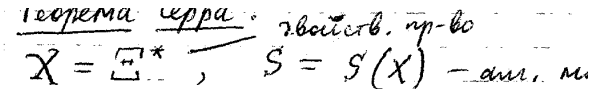
(= quotient sheaves of the sheaves of sections of bundles).

This is more convenient, since one can take images, cokernels, etc..

(~ adjoining differences).

Serre's theorem:

$$X = \Xi^*, \qquad S = S(X) \text{ — the algebra of polynomials.}$$



Original source details: GELFAND-P000759

Diagram and typed description.

Ξ^* / dual space

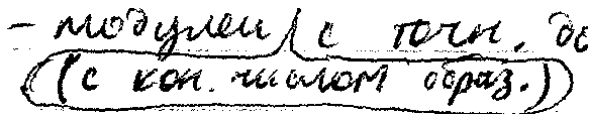
The category of sheaves on $\mathbb{P}^n$ (coherent) [crossed out: po?]



Uncertain reading. The cancelled word before "эквивал." (equivalent) tentatively reads "по", possibly "до".

Original source details: GELFAND-P000759

is equivalent to the category of graded $S(X)$-modules up to finite-dimensional ones.



Original source details: GELFAND-P000759

Diagram and typed description.

modules

(with finitely many generators)

$$\mathcal{E} \longmapsto M = \bigoplus M_j, \qquad M_j = \Gamma(\mathcal{E} \otimes O(j), \mathbb{P}^n)$$

$$S_k = \Gamma(O(k), \mathbb{P}^n)$$

The module structure is introduced in the obvious way.



Diagram and typed description.

Original source details: GELFAND-P000759

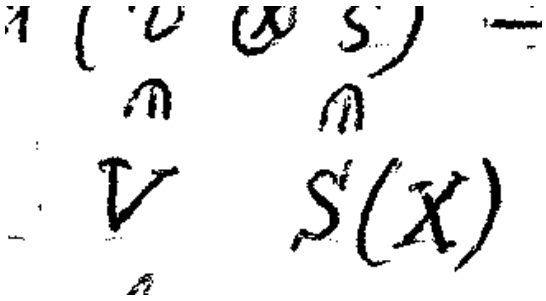
Proof of Theorem 1.

1°. From a Λ-module V construct a complex of S-modules $W = (W^k, d)$:

$$W^k = V_{-k} \otimes_{\mathbb{C}} S(X)$$

Let $x_0, \ldots, x_n$ — a basis in X, dual to the basis in Ξ. set

$$d(v \otimes s) = \sum \xi_i v \otimes x_i s.$$



Original source details: GELFAND-P000759

Diagram and typed description.

$$\begin{array}{cc} v & s \\ \cup & \cup \\ V & S(X) \end{array}$$

It is easy to check that $d : W^k \to W^{k+1}$ and $d^2 = 0$.

In each W^k define the grading:

$$(W^k)_j = V_{-k} \otimes S_{j+k}$$

d — a morphism of graded modules (it does not change degree).

GELFAND-P000760

(This is the algebraic version of the geometric construction given above)

$V \longmapsto$ the complex of bundles $\mathcal{L}(V)$

$\overline{W}$ — This is the algebraic analogue of $\mathcal{L}(V)$.



Original source details: GELFAND-P000760

Uncertain reading. A short detached stroke over the left of W is represented by an overbar.

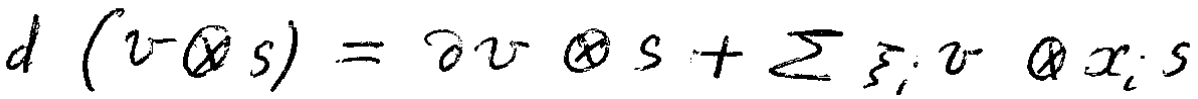
Generalization: consider a complex of Λ-modules

$$\ldots \xrightarrow{\partial} V^i \xrightarrow{\partial} V^{i+1} \xrightarrow{\partial} V^{i+2} \xrightarrow{\partial} \ldots$$

Imprecisely: construct $W^{i,k}$ (bicomplex) and sum over the diagonals
(the complex from the bicomplex)

$$V^i = \bigoplus (V^i)_j; \qquad W^k_j = \bigoplus_\ell V^{\ell+k}_{j-\ell} \otimes S_\ell$$

$$d(v \otimes s) = \partial v \otimes s + \sum \xi_i v \otimes x_i s$$



Original source details: GELFAND-P000760

Uncertain reading. There is a plus sign between the terms. No additional sign factor is visible before the sum.

It does not change [crossed out: its?]



Uncertain reading. The short cancelled entry may read “ero” (its) or j.

Original source details: GELFAND-P000760

the lower index, and increases the upper one by 1.

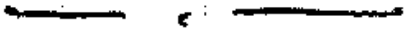


Diagram and typed description.

——— • ———

Original source details: GELFAND-P000760

2°. Complex of graded S-modules $\longmapsto$ complex of graded Λ-modules

$$\ldots \longrightarrow W^k \xrightarrow{d} W^{k+1} \xrightarrow{d} \ldots$$

Set $\overline{W} = \bigoplus W^k$



Uncertain reading. Here W is underlined and also has a short stroke over its left side.

Original source details: GELFAND-P000760

and $\underline{V} = \mathrm{Hom}_{\mathbb{C}}(\Lambda, W)$

Introduce a grading on $\underline{V}$:

$$v \in V_j^i \Longleftrightarrow v(\Lambda_k) \subset W_{j+k}^{i-j-k} \quad \text{for all } k.$$

Set:

$$\partial : V^i \longrightarrow V^{i+1}$$

$$\partial(v)(\lambda) = d(v(\lambda)) + \sum x_i v(\xi_i \lambda)$$

This map is in a certain sense inverse to the preceding one.

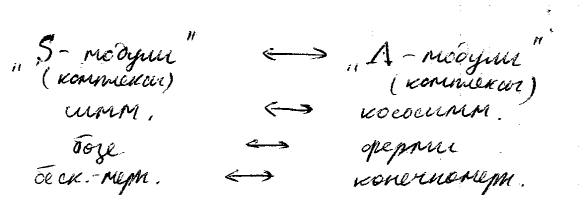


Diagram and typed description.

"S-modules" (complexes)	$\longleftrightarrow$	"Λ-modules" (complexes)
symm.	$\longleftrightarrow$	skew-symm.
Bose	$\longleftrightarrow$	Fermi
infinite-dim.	$\longleftrightarrow$	finite-dim.

Original source details: GELFAND-P000760

2°. $C^b(\Lambda)$ — the category of finite-length complexes of Λ-modules.

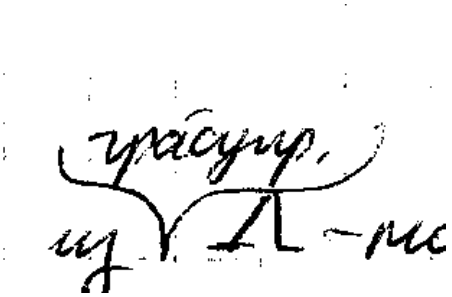


Diagram and typed description.

graded

of Λ-modules

Original source details: GELFAND-P000760

$C^b(S)$ — __ " ___ of finitely generated graded S-modules

(the differential preserves the grading)

GELFAND-P000761

[source number: 57]

$C(\Lambda)$ — the category of infinite complexes of graded Λ-modules (but infinite-dimensional ones)

The functors have been constructed:

$$F : \ C^b(\Lambda) \longrightarrow C^b(S)$$
$$V \longmapsto W \qquad (\text{see item } 1^\circ).$$



Uncertain reading. The upper left stroke touches W and is rendered as part of the letter.

Original source details: GELFAND-P000761

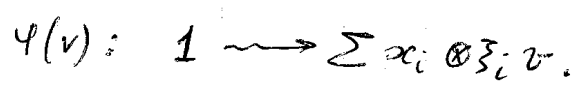
$$G:\ C^b(S) \longrightarrow C(\Lambda) \qquad \text{(see item } 2^\circ\text{)}.$$

Main technical theorem.

Consider the natural inclusion $i : C^b(\Lambda) \longrightarrow C(\Lambda)$. There is a morphism of functors $i \longrightarrow G \circ F$, inducing an isomorphism in cohomology.

This means that there is a natural map

$$\varphi : V \rightsquigarrow \mathrm{Hom}(\Lambda, S \otimes V)$$
$$\varphi(v) : \quad 1 \rightsquigarrow \sum x_i \otimes \xi_i v.$$



Uncertain reading. The manuscript gives the value at 1 as $\sum x_i \otimes \xi_i v$.

Original source details: GELFAND-P000761

($\forall V$ one must construct a homomorphism



Undeciphered passage. A sign after the abbreviation for homomorphism is heavily crossed out.

Original source details: GELFAND-P000761

$V \longrightarrow G \circ F(V)$)

V — a one-dimensional trivial module $\Rightarrow$ on the right is the Koszul complex.



Diagram and typed description.

——— • ———

Original source details: GELFAND-P000761

A — a category (abelian), $C(A)$ — the category of complexes over A (infinite ones)

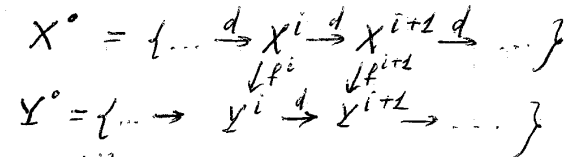


Diagram and typed description.

$$X^\bullet = \{\dots \xrightarrow{d} X^i \xrightarrow{d} X^{i+1} \xrightarrow{d} \dots\}$$
$$\downarrow f^i \qquad \downarrow f^{i+1}$$
$$Y^\bullet = \{\dots \longrightarrow Y^i \xrightarrow{d} Y^{i+1} \longrightarrow \dots\}$$

Original source details: GELFAND-P000761

morphism $\{f^i\}$ — the diagram commutes.

Cohomology:

$$H^i(X^\bullet) = \ker d / \operatorname{Im} d$$

map

$$X^\bullet \xrightarrow{f} Y^\bullet \rightsquigarrow f_i^* : H^i(X^\bullet) \longrightarrow H^i(Y^\bullet)$$

f is called a quasi-isomorphism, if it induces an isomorphism on all cohomology groups.

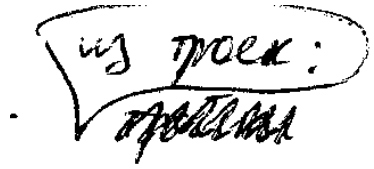
Roughly: the derived category is obtained by inverting quasi-isomorphisms.

Precise definition.

S — the set of quasi-isomorphisms. ("ensemble" in Verdier)

$$\mathrm{Ob}(D(A)) = \mathrm{Ob}(C(A))$$

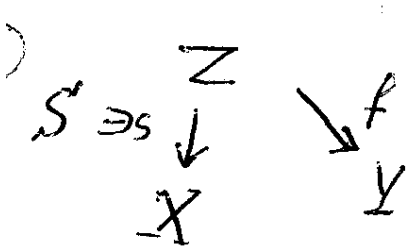
$\mathrm{Hom}_{D(A)}(X, Y)$ — [crossed out: triples?]



Original source details: GELFAND-P000761

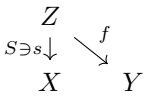
Uncertain reading. The crossed-out word tentatively reads "тройки" (triples). Above it the outlined insertion reads "из троек:" (from triples).

[inserted above in an outline: from triples:]



Original source details: GELFAND-P000761

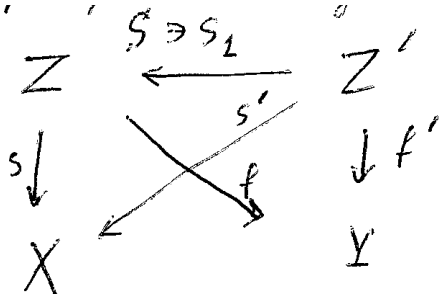
Diagram and typed description.



GELFAND-P000762

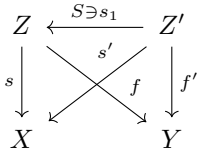
[Crossed out: 2 triples are called equivalent if]

Map of triples: a commutative diagram.



Original source details: GELFAND-P000762

Diagram and typed description.



$\mathrm{Hom}_{D(A)}(X, Y)$ — the inductive limit over such triples.

Element:

$$Z \longleftarrow Z' \longleftarrow Z'' \longleftarrow \cdots$$

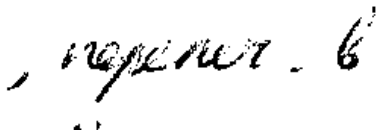
identified if they agree from some point onward

2 places where this is set out:

1) Hartshorne 20 vol. L.N.

Residues and duality

2) Verdier dissertation abstract (1963), reprinted?



Original source details: GELFAND-P000762

Uncertain reading. The abbreviation between the date and the volume number tentatively reads “перепеч.” (reprinted).

in 569 vol. Lect.Notes

(there are no proofs here)

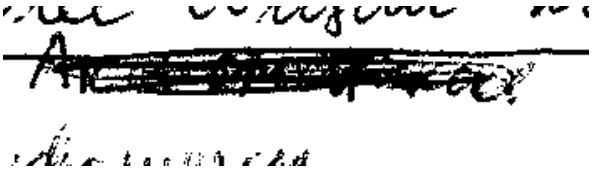
More general construction (axiomatic)



Original source details: GELFAND-P000762

Diagram and typed description. —— . ——

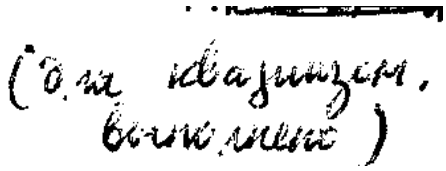
[Crossed out: A...]



Original source details: GELFAND-P000762

Undeciphered passage. The initial A is visible in the cancelled heading. Dense strokes obscure the remainder.

[In the left margin: (for quasi-isomorphisms they hold)]



Original source details: GELFAND-P000762

Uncertain reading. The end of the abbreviated word “квазизм.” (quasi-isomorphisms) in the margin is tentative.

1)

$$\left.\begin{array}{l} s_1 \in S, \quad s_2 \in S \\ s_1 s_2 \text{ — exists} \end{array}\right\} \Rightarrow s_1 s_2 \in S.$$

2)

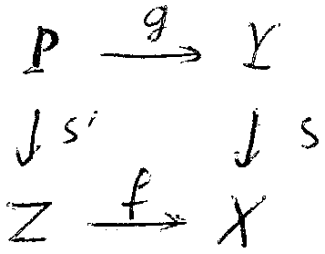


Diagram and typed description.

$$\begin{array}{ccc} P & \xrightarrow{g} & Y \\ \downarrow s' & & \downarrow s \\ Z & \xrightarrow{f} & X \end{array}$$

Original source details: GELFAND-P000762

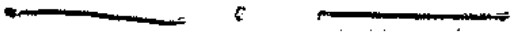
$$\forall\ Z, X, Y,\ f, s$$
$$\exists\ P, g, s'$$

3) Let $f : X \longrightarrow Y$. Then the following properties are equivalent

(i) $\exists\ s \in S :\ sf = 0$

(ii) $\exists\ s' \in S :\ fs' = 0$

Diagram and typed description.



Original source details: GELFAND-P000762

What is needed to construct the morphism of functors?

(the one we need).

1) It is constructed.

2) One must check that for one-dimensional V everything is in order (simple calculations with the Koszul complex)

3) Introduce filtrations, the factors are standard.

Derived category: reduction to the factors.

GELFAND-P000763

[source number: 62]

6/III–78 Gelfand seminar.

[I. M. G.] Sheaf of C^∞ functions on $(-1, 1)$, vanishing at 0 at the point 0 — a sheaf of sections of a vector bundle?

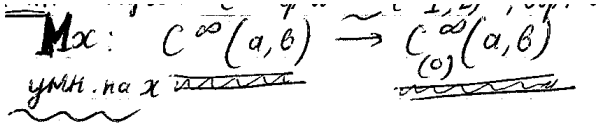


Diagram and typed description.

$$\begin{array}{ccc} Mx: & C^\infty(a,b) & \longrightarrow & C^\infty_{(0)}(a,b) \\ & \text{mult. by } x & & \end{array}$$

Original source details: GELFAND-P000763

isomorphism of $C^\infty(a,b)$-modules.



Diagram and typed description. Aomoto

Original source details: GELFAND-P000763

Exactly solvable models in neutral massive field theories. (on $\mathbb{R}^2$)



Uncertain reading. The final letter x of «нейтральных» (neutral) is retraced.

Original source details: GELFAND-P000763

3 types of fields.

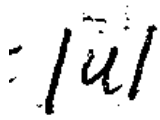
1°. Auxiliary fermions (on $\mathbb{R}$)

on $\mathbb{R}$; $u \in \mathbb{R}$; fields covariant with respect to the action of $\mathbb{R}^+$ on $\mathbb{R}$;

$\psi(u)^+$ and $\psi(u)$ — creation and annihilation operators;

$$\psi(-u) = \psi^+(u); \tag{1}$$

$$[\psi(u), \psi(u')]_+ = 2\pi|u|\delta(u+u'); \tag{2}$$



Uncertain reading. Short separate strokes are visible above the u in the absolute value. Their status as a distinct sign is uncertain.

Original source details: GELFAND-P000763

left-hand side $\underset{\text{def}}{=} \psi(u)\psi(u') + \psi(u')\psi(u)$,

2°. $\varphi_F(x)$, $\varphi^F(x)$. (on $\mathbb{R}^2$)

Consider a pair of functions:

$$w(x) = (w_+(x), w_-(x))$$

$$\frac{\partial w_\pm}{\partial x^\pm} \mp m w_\mp(x) = 0, \tag{3}$$

(Dirac equation) (2 equations), m — a number.

where

$$x^\pm = \frac{x^0 \pm x^1}{2}, \qquad p^\pm = p^0 \pm p^1.$$

$$\langle w, w' \rangle = \int_{-\infty}^{\infty} dx^1 \big(w_+(x) w'_+(x) + w_-(x) w'_-(x)\big) \tag{4}$$

$$\psi_\pm(x) = \int_{-\infty}^{\infty} \underline{du}\, \sqrt{0 + iu}^{\pm} \psi(u) \exp\left(-im\left(x^- u + \frac{x^+}{u}\right)\right), \tag{5}$$



Uncertain reading. A small oval is visible above the right parenthesis. Its meaning is uncertain.

Original source details: GELFAND-P000763

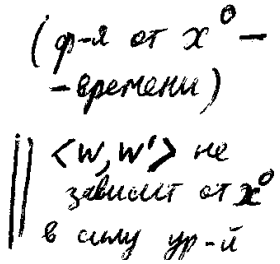


Diagram and typed description.
(a function of x^0 — time)
$\| \langle w, w' \rangle$ is independent of x^0 by the equations.

Original source details: GELFAND-P000763

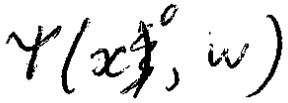
Here

$$\underline{du} = \frac{du}{2\pi|u|}$$

GELFAND-P000764

$\pm$ above the radical — choice of branch $+0$ or -0

$$w(x) \longleftrightarrow \int_{-\infty}^{\infty} dx^1 (w_+\psi_+ + w_-\psi_-) = \psi(x^0, w) \tag{6}$$



Uncertain reading. The first argument contains a dense correction after x, with a visible 0 above it. The reading x^0 is tentative.

Original source details: GELFAND-P000764

This is already an operator

There is an orthogonal decomposition of the solution space:

$$W = \left\{ w \;\middle|\; \frac{\partial w_{\pm}}{\partial x^{\pm}} \mp m w_{\mp} = 0 \right\} = W_x^+ \oplus W_x^-, \tag{7}$$

where

$$W_x^+ = \{ w \in W \mid w = 0 \text{ in the hatched region} \} \tag{8}_1$$

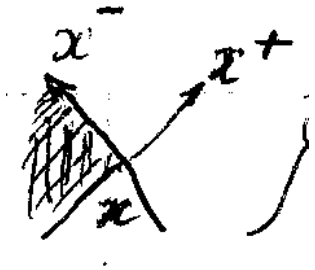
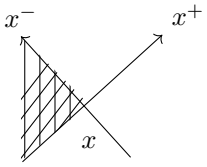


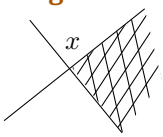
Diagram and typed description.



Original source details: GELFAND-P000764

$$W_x^- = \{ \dots \mid w = 0 \text{ in the hatched region} \}. \tag{8}_2$$

Diagram and typed description.





Original source details: GELFAND-P000764

In the Bose case instead of the Fermi case, the functions are analogous to a Lagrangian decomposition into p and q (into 2 Lagrangian planes).

$A(W)$ — [Crossed out: ...] Clifford algebra over W

Undeciphered passage. Slanted strokes obscure the short crossed-out entry before “Clifford”.



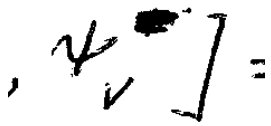
Original source details: GELFAND-P000764

$$\{\psi_\mu^\pm\} \quad \text{— bases in } W_x^\pm$$

The Clifford algebra is generated by them with relations

$$\psi_\mu^+\psi_\nu^- + \psi_\nu^-\psi_\mu^+ \underset{\text{def}}{=} [\psi_\mu^+, \psi_\nu^-] = \delta_{\mu\nu} \tag{9}_1$$

$$[\psi_\mu^+, \psi_\nu^+] = [\psi_\mu^-, \psi_\nu^-] = 0 \tag{9}_2$$



Uncertain reading. The superscript of the final ψ_ν is heavily crossed over. The minus is tentative.

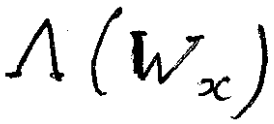
Original source details: GELFAND-P000764

Notation:

$$\underline{\underline{\psi_\mu^- = \psi_\mu}}$$

Relation:

$$A(W) = \Lambda(W_x^+) \cdot \Lambda(W_x) \tag{10}$$



Uncertain reading. The second factor $\Lambda(W_x)$ shows the subscript x and has no distinct superscript.

Original source details: GELFAND-P000764

The bases are orthogonal with respect to the scalar product (4).

There is a map:

$$A(W) \longrightarrow \Lambda W_x^+ \wedge \Lambda W_x^- \qquad \text{isomorphism}$$

$$\lambda \cdot \mu \overset{N}{\longrightarrow} \lambda \wedge \mu$$

inverse map:

$$:\varphi: \longleftarrow \varphi$$

(Wick product)

In the Bose case this is the “symbol of the operator”.

GELFAND-P000765

[source number: 63]

Definition.

$$\varphi_F(x) \in A(W)$$

$$\text{Spin}(W_x) = \{g \in A(W_x) \mid gW_xg^{-1} \subset W_x\}$$

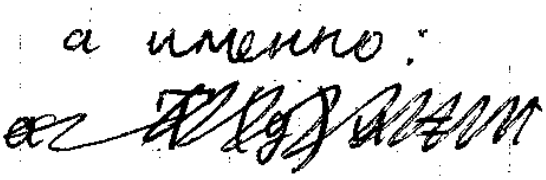
In fact

$$\varphi_F(x) \in \text{Spin}(W)$$

There is a map:

$$\text{Spin}(W) \xrightarrow{T} O(W); \qquad \text{namely} \quad T_g w = gwg^{-1}; \tag{11}$$

[Crossed out: and ...]



Undeciphered passage. Between $O(W)$ and $T_g w$, "a" (and) and a mark possibly reading $T(g)$ are visible. Dense crossings obscure the rest. "Namely" is written above.

Original source details: GELFAND-P000765

W has a scalar product and hence we have a definition of $O(W)$.

$$T_{\varphi_F(x)}(w^+ + w^-) \underset{\text{def}}{=} w^+ - w^- \tag{12}$$

(this is the definition of $\varphi_F(x)$).

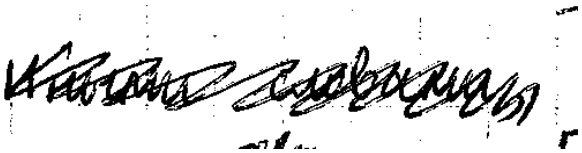
T fails to be a monomorphism, so the definition is still ill-defined. More precisely,

$$\varphi_F(x) = :\exp L_F(x): \tag{13}$$

where

$$\Lambda W \ni L_F(x) = \frac{1}{2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \underline{du}\,\underline{du'}\,\frac{-i(u-u')}{u+u'-i0}\cdot\Psi(u)\Psi(u') \exp\left(-im\left(x^-(u+u') + x^+\left(\frac{1}{u}+\frac{1}{u'}\right)\right)\right). \tag{14}$$

[Crossed out: One can speak?]

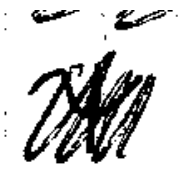


Uncertain reading. Two words beneath (14) are heavily crossed out. "Можно говорить" (one can speak) is tentative.

Original source details: GELFAND-P000765

Next,

[Crossed out: ...]



Undeciphered passage. Dense slanted strokes obscure the mark above and to the left of (15).

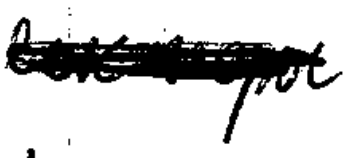
Original source details: GELFAND-P000765

$$\varphi^F(x) = \Psi_0(x) \exp L_F(x), \tag{15}$$

where

$$W \ni \Psi_0(x) = \int_{-\infty}^{\infty} \underline{du}\, \Psi(u) \exp\left(-im\left(x^- u + \frac{x^+}{u}\right)\right) \tag{16}$$

There are [Crossed out: vectors?] 2 objects:



Original source details: GELFAND-P000765

Uncertain reading. The crossed-out word before “2 objects” tentatively reads “векторы” (vectors).

$$\langle \text{vac}|, \qquad |\text{vac}\rangle$$

$$A(W)/A(W)W_x^- = |\text{vac}\rangle \tag{17}_1$$

$$W_x^+ A(W)\backslash A(W) = \langle \text{vac}| \tag{17}_2$$

Properties:

$$W_x^-|\text{vac}\rangle = 0, \qquad \langle \text{vac}|W_x^+ = 0 \tag{18}$$

(natural for the vacuum).

Consider the function

$$\langle \text{vac}|\Psi(x_1)\dots\Psi(x_k)\varphi_F(a_1)\dots\varphi_F(a_m)|\text{vac}\rangle; \tag{19}$$

($\Psi(x)$ from formula (6)).

defined on $(\mathbb{R}^2)^k$

All a_ν are fixed.

(2-dimensional Ising model?)

$k = 1 \quad \Rightarrow$

$$\langle \text{vac}|\Psi(x)\varphi_F(a_1)\cdot\varphi^F(a_i)\dots\varphi_F(a_n)|\text{vac}\rangle \underset{\text{def}}{=} W^i_{F,n}(x) \tag{20}$$

A system of equations is obtained; put

$$M_F = z\partial_z - \bar{z}\partial_{\bar{z}} + \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \tag{21}$$

The 1st equation has the form:

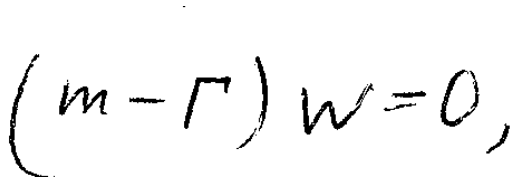
$$M_F w = (Bm^{-1}\partial_z - \bar{B}m^{-1}\partial_{\bar{z}} + E)w \tag{22}$$

$$z = (x^0 + x^1), \qquad \bar{z} = (x^0 - x^1)$$

B, E — constant matrices

The 2nd equation

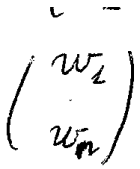
$$(m - \Gamma)w = 0, \qquad \Gamma = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \tag{23}$$



Uncertain reading. The manuscript writes $m-$ before Γ in (23).

Original source details: GELFAND-P000766

$$w_i = \begin{pmatrix}w_{i+}\\ w_{i-}\end{pmatrix}, \qquad \begin{pmatrix}w_1\\ w_n\end{pmatrix}$$



Uncertain reading. The subscript of the lower entry in the right column reads n.

Original source details: GELFAND-P000766

(22)–(23) — a maximally overdetermined system
(in fact this is the Gauss–Manin connection).
Consider also

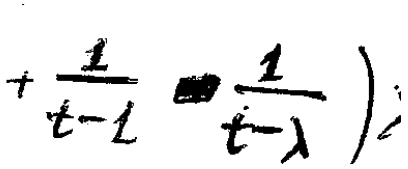
$$\langle \text{vac}|\varphi_F(a_1)\dots\varphi_F(a_n)|\text{vac}\rangle = \tau \tag{24}$$

as a function of a. (τ-function)
For $n = 2$ this is a Painlevé function of the 3rd kind.
[Written beneath “Пенлеве”: (Painleve)]
Wu, Mc Coy, Tracy, Phys. Rev. B 13 (1976)
[Beneath the names] something about Painlevé functions.
I. M. G.
Garnier — a memoir on Painlevé functions.
6 types of equations, 2nd order, fixed singular points. The most general

$$\begin{aligned}\frac{1}{y}y'' = {} & \frac{a}{x^2} + \frac{b}{(x-1)^2} + \frac{c}{(x-t)^2} + \frac{d}{x(x-1)} + \frac{3}{4(x-\lambda)^2} \\ & + \frac{\alpha}{x(x-1)(x-t)} + \frac{\beta}{x(x-1)(x-\lambda)}\end{aligned} \tag{А}$$

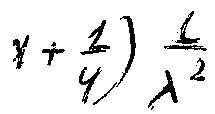
The periods (monodromy) are independent of the parameters $\Leftrightarrow$

$$\begin{aligned}\Leftrightarrow \quad \lambda'' = {} & \frac{1}{2}\left(\frac{1}{\lambda} + \frac{1}{\lambda-1} + \frac{1}{\lambda-t}\right)(\lambda')^2 \\ & - \left(\frac{1}{t} + \frac{1}{t-1}\,[?]\,\frac{1}{t-\lambda}\right)\lambda' + \\ & + 2\frac{\lambda(\lambda-1)(\lambda-t)}{t^2(t-1)^2}\left[a + b + c + d + 1 - \left(a + \frac{1}{4}\right)\frac{1}{\lambda^2}\right. \\ & \left. + \left(b + \frac{1}{4}\right)\frac{t-1}{(\lambda-1)^2} - \frac{ct(t-1)}{(\lambda-t)^2}\right]\end{aligned} \tag{Б}$$



Original source details: GELFAND-P000766

Uncertain reading. A dense correction obscures the sign between $1/(t-1)$ and $1/(t-\lambda)$.



Original source details: GELFAND-P000766

Uncertain reading. The manuscript writes 1 over the first λ^2.

Only these have fixed singular points.
Constancy of the matrix monodromy (А) $\Leftrightarrow$ fixedness of the singular points of (Б)
t, λ vary in accordance with (Б)

GELFAND-P000767



Diagram and typed description. 13/III – 78 Gelfand Seminar

Original source details: GELFAND-P000767

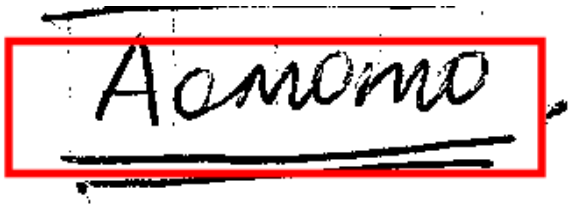


Diagram and typed description. Aomoto.

Original source details: GELFAND-P000767

On generalizing the classical hypergeometric function using integrals of the type of Bernstein (I.)



Uncertain reading. "И." (I.) appears in parentheses after "Bernstein". The initial is left unexpanded.

Original source details: GELFAND-P000767

Appell function.
The case of 2 variables.

I. M. G.
Classical hypergeometric function:

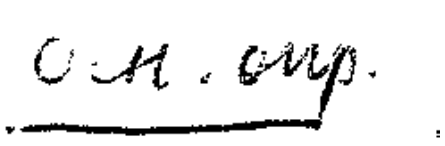
$$\sum_{n=0}^{\infty} x^n \frac{\alpha(\alpha+1)\dots(\alpha+n-1)\,\beta(\beta+1)\dots(\beta+n-1)}{\gamma(\gamma+1)\dots(\gamma+n-1)\,n!}$$

GELFAND-P000768

$$F_1 = \sum_{m,n\geq 0} \frac{(\alpha, m+n)(\beta, m)(\beta', n)}{(\gamma, m+n)(1, m)(1, n)} x^m y^n \tag{1}$$

— · —

I. M. G. One? definition of hypergeometric functions:



Uncertain reading. The first abbreviation before "опр." (definition) tentatively reads "Од." (one).

Original source details: GELFAND-P000768

$$\left((x^2-1)^n\right)^{(n)} \qquad \text{(Legendre polynomials)}$$
$$\|$$
$$\left[((x-1)(x+1))^n\right]^{(n)}$$

Generalization:

$$\left[(x-1)^{\alpha-1}(x+1)^{\beta-1}\right]^{(\gamma)}$$

Integrals of order α.

$$I_\alpha f = \int_0^x \underbrace{\frac{(x-t)^{\alpha-1}}{(\alpha-1)!}}_{\Gamma(\alpha)} f(t)\,dt, \qquad D^\alpha = I_{-\alpha} \quad \text{(analytic continuation)}$$

$$\left[x^{\alpha-1}(1-x)^{\beta-1}\right]^{(\gamma)}$$

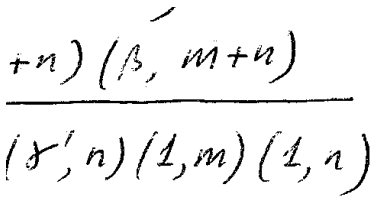
Hypergeometric function: 1) good series expansion 2) good differential equation 3) good integral.

$$\int_0^x t^{\alpha-1}(1-t)^{\beta-1}(x-t)^{\gamma-1}\,dt$$

B-functions are simpler.

— · —

$$(\alpha, n) = \alpha(\alpha+1)\dots(\alpha+n-1)$$
$$F_4 = \sum_{m,n\geq 0} \frac{(\alpha, m+n)(\beta, m+n)}{(\gamma, m)(\gamma', n)(1, m)(1, n)} \tag{2}$$



Uncertain reading. The factor $x^m y^n$ is absent after the fraction in (2).

Original source details: GELFAND-P000768

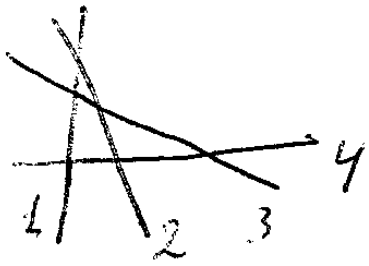
There is an integral representation [Crossed out: ...] for F_1



Undeciphered passage. Slanted strokes obscure the short mark before "for F_1".

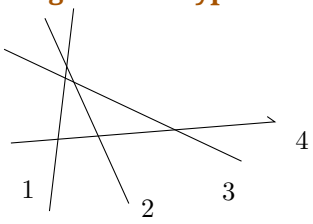
Original source details: GELFAND-P000768

(3)



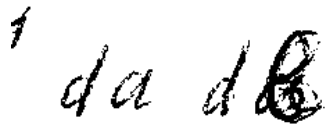
Original source details: GELFAND-P000768

Diagram and typed description.



Line 4 has a short hook at its right end. Its function is uncertain.

$$\int f_1^{\lambda_1} f_2^{\lambda_2} f_3^{\lambda_3} f_4^{\lambda_4}\, da\, db$$



Original source details: GELFAND-P000768

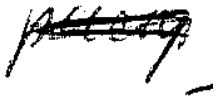
Uncertain reading. The loops in db overlap. The b is tentative.

f_j — linear forms; x, y are among the coefficients.

3 lines are reduced to a simple form (affine transformation) and fixed

The 4th is given by parameters a, b.

F_2, F_3 are obtained similarly, but with a different [Crossed out: arrangement?] arrangement of lines, (out of general position):



Original source details: GELFAND-P000768

Uncertain reading. The crossed-out word tentatively begins “расп” (arrangement).

[Crossed out: F_1?]



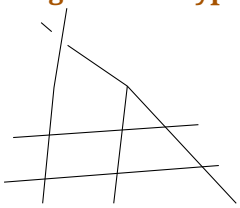
Original source details: GELFAND-P000768

Uncertain reading. The crossed-out mark at lower left resembles F with a short stroke at its lower right. F_1 is tentative.



Original source details: GELFAND-P000768

Diagram and typed description.



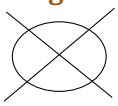
GELFAND-P000769

[source number: 68]

F_4 :



Diagram and typed description.



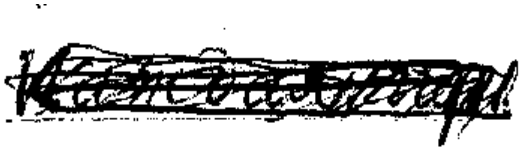
Original source details: GELFAND-P000769

a quadratic form is used in place of 2 lines.

The number of linearly independent solutions of the corresponding differential equations = the number of bounded connected components of the corresponding picture.

General remark on the coefficients: the ratio of neighbors = a rational function of m, n.

[Crossed out: ...]



Undeciphered passage. Dense horizontal and slanted strokes obscure the opening words before “An analogous problem”.

Original source details: GELFAND-P000769

An analogous problem can be considered with any number of hyperplanes

$$f_i = \sum_{\nu=1}^{n} a_{i\nu} x_\nu + a_{i0} \tag{4}$$

$$1 \le i \le m$$

$f_{m+1} = 1$ Usually $m \ge n$.

Consider the matrix formed from the coefficients of the forms

$$A = \begin{pmatrix} a_{10} & \dots & a_{1n} \\ \vdots & & \\ a_{m0} & \dots & a_{mn} \\ 1 & \dots\ 0 & 0 \end{pmatrix} \in \mathbb{C}^{m(n+1)} \tag{5}$$

The affine group in n variables acts there

(on x it acts)

— Change of variable in x

Let

$$\mathcal{M} \subset \mathbb{C}^{m(n+1)} \times \mathbb{C}^n$$

consist of pairs

$$\{(A, x) : f_j(x) \ne 0\} \qquad \text{(none of the equations is satisfied)}$$

(complement of all the hyperplanes)

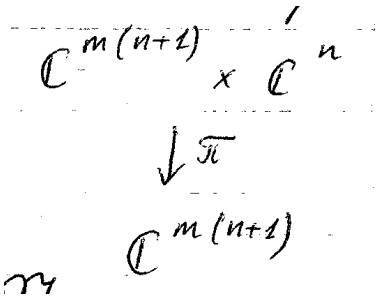


Diagram and typed description.

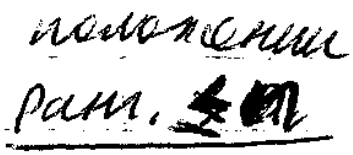
$$\begin{array}{c} \mathbb{C}^{m(n+1)} \times \mathbb{C}^n \\ \downarrow \pi \\ \mathbb{C}^{m(n+1)} \end{array}$$

Original source details: GELFAND-P000769

$$\gamma \subset \mathcal{M}$$

the set of such equations for which the hyperplanes are out of general position

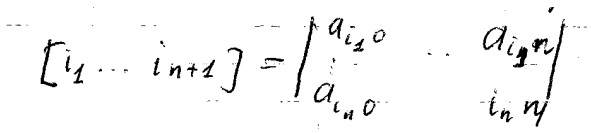
rank $\le n$



Uncertain reading. The inequality after “rank” is densely overwritten. The reading $\le n$ is tentative.

Original source details: GELFAND-P000769

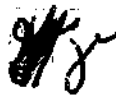
$$[i_1 \dots i_{n+1}] = \begin{vmatrix} a_{i_1 0} & \dots & a_{i_1 n} \\ \vdots & & \\ a_{i_n 0} & & i_n n \end{vmatrix} \qquad \text{(minor obtained from rows } i_1, \dots, i_n\text{)}$$



Uncertain reading. The left bracket has i_{n+1}, while the last row and explanation have i_n. The lower right entry is written $i_n n$.

Original source details: GELFAND-P000769

The projection of [Crossed out: γ?] γ is given by conditions on minors:



Uncertain reading. The looped crossed-out sign before the clean γ may be γ.

Original source details: GELFAND-P000769

$$Y = \{A \mid [i_1, \dots, i_{n+1}] = 0\} \tag{6}$$

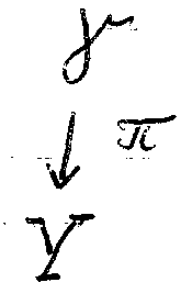


Diagram and typed description.

$$\begin{array}{c} \gamma \\ \downarrow \pi \\ Y \end{array}$$

Original source details: GELFAND-P000769



Uncertain reading. A separate densely retraced sign beneath (6), at the left, tentatively reads γ.

Original source details: GELFAND-P000769

GELFAND-P000770

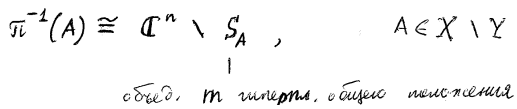


Diagram and typed description.

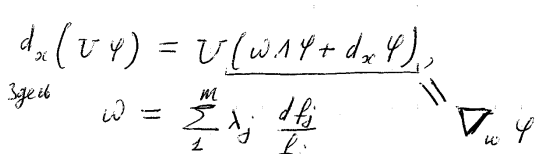
$$\pi^{-1}(A) \cong \mathbb{C}^n \setminus S_A \quad , \qquad A \in X \setminus Y$$

union of m hyperplanes in general position

Original source details: GELFAND-P000770

$\Omega_A^\bullet$ — the complex of rational forms in $\mathbb{C}^n$ with poles on S_A
(exterior differentiation)
$\forall \varphi \in \Omega_A$ there is a formula

$$U = f_1^{\lambda_1} \dots f_m^{\lambda_m},$$



Original source details: GELFAND-P000770

Diagram and typed description.

$$d_x(U\varphi) = U\underbrace{(\omega \wedge \varphi + d_x\varphi)}_{=\nabla_\omega \varphi},$$

Here

$$\omega = \sum_1^m \lambda_j \frac{df_j}{f_j} \qquad (\nabla_\omega \text{ — a connection}).$$

ω — the connection form

$$\nabla_\omega : \Omega_A^p \longrightarrow \Omega_A^{p+1}$$
$$\nabla_\omega \varphi = \omega \wedge \varphi + d_x \varphi$$

(a connection in the trivial bundle)

We have:

$$d\omega + \omega \wedge \omega = 0$$

(integrability condition)

It follows that $\nabla_\omega^2 = 0$ and we obtain a complex.

Deligne, Grothendieck: a theory of rational de Rham hypercohomology arose.

Assumption.

$$\lambda_{m+1} = -\lambda_1 - \ldots - \lambda_m$$

$\forall$ collections

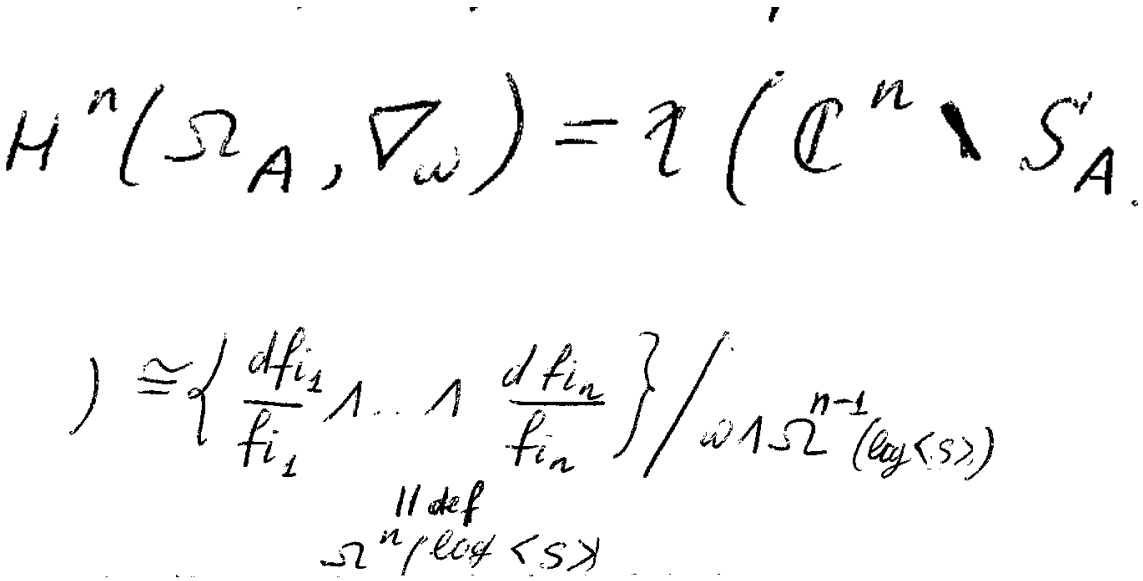
$$1 \le i_1 < \ldots < i_\nu \le m+1, \qquad 1 \le \nu \le n.$$

Condition:

$$\lambda_{i_1} + \ldots + \lambda_{i_\nu} \notin \mathbb{Z}^+ \qquad \mathbb{Z}^+ = \{n \in \mathbb{Z},\ n > 0\}$$

Proposition.

$$H^p(\Omega_A, \nabla_\omega) = 0, \quad p \ne n$$
$$H^n(\Omega_A, \nabla_\omega) = \chi(\mathbb{C}^n \setminus S_A)$$
$$\left\{ \frac{df_{i_1}}{f_{i_1}} \wedge \ldots \wedge \frac{df_{i_n}}{f_{i_n}} \right\}$$
$$\cong \frac{\Omega^n(\log\langle S\rangle)}{\omega \wedge \Omega^{n-1}(\log\langle S\rangle)} \quad = \text{def}$$



Original source details: GELFAND-P000770

Uncertain reading. The manuscript writes $H^n(\Omega_A, \nabla_\omega) = \chi(\mathbb{C}^n \setminus S_A)$, with no dim before H^n.

GELFAND-P000771

[source number: 69]

Explanation:

$$\Omega^1(\log\langle S\rangle) = \left\{\frac{df_i}{f_i}\right\}_{i\le m}$$

$$\bigwedge \Omega^1(\log\langle S\rangle)/\sim\ = \Omega^\bullet(\log\langle S\rangle)$$

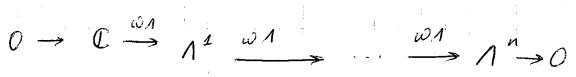


Diagram and typed description.
$0 \longrightarrow \mathbb{C} \overset{\omega\wedge}{\longrightarrow} \Lambda^1 \overset{\omega\wedge}{\longrightarrow} \cdots \overset{\omega\wedge}{\longrightarrow} \Lambda^n \longrightarrow 0$

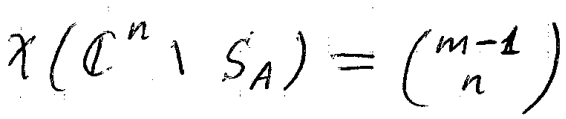
Original source details: GELFAND-P000771

(They are closed: $\nabla_\omega = \omega\wedge$ for such forms)

$$\chi(\mathbb{C}^n \setminus S_A) = \binom{m-1}{n}$$

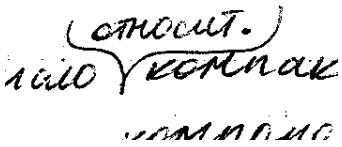
$$= \text{the number of (relatively) compact connected}$$

components of $\mathbb{R}^n \setminus S_A$



Uncertain reading. The manuscript has no additional sign factor before $\binom{m-1}{n}$.

Original source details: GELFAND-P000771



Uncertain reading. “(относит.)” (relatively) is inserted above the line before “compact”.

Original source details: GELFAND-P000771

—, —

$\int U\varphi$ [Crossed out: ...] Duality

$\varphi \in \Omega_A$



Undeciphered passage. The short entry between the integral and “Duality” is heavily crossed out.

Original source details: GELFAND-P000771

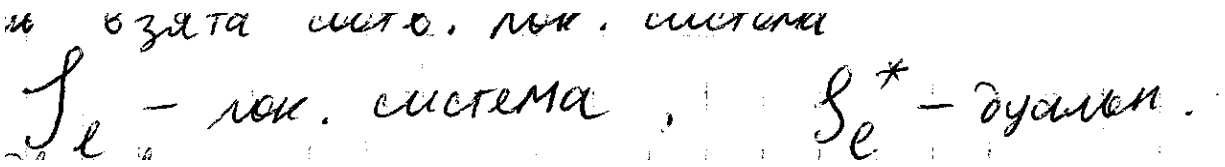
U is multivalued $\Rightarrow$ continuation around loops gives a representation of the fundamental group:

$$\ell_U : \pi_1(\mathbb{C}^n - S_A) \longrightarrow \mathbb{C}^*$$

(continuation multiplies it by a nonzero complex number)

Take the corresponding local system

$$\zeta_\ell \text{ — local system,} \qquad \zeta_\ell^* \text{ — the dual.}$$



Original source details: GELFAND-P000771

Uncertain reading. The looped local system symbol with subscript ℓ is tentatively rendered as ζ_ℓ.

There is a duality

$$H^n(\mathbb{C}^n \setminus S_A, \zeta_\ell) \times H_n(\mathbb{C}^n - S_A, \zeta_\ell^*) \longrightarrow \mathbb{C}$$

$\forall$ compact connected component $\longrightarrow$ element $\in$

$$H_n(\mathbb{C}^n - S_A, \zeta_\ell)$$



Original source details: GELFAND-P000771

Uncertain reading. The final local system symbol is heavily retraced. A separate star is indistinct, unlike in the preceding pairing.

GELFAND-P000772

20/III–78 Gelfand Seminar

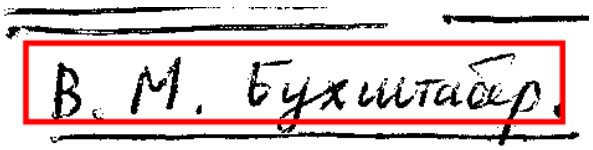
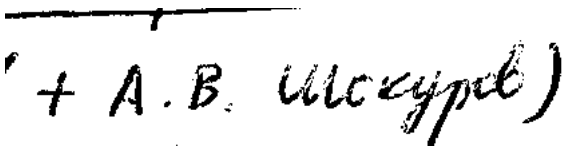


Diagram and typed description.

V. M. Buchstaber.

Original source details: GELFAND-P000772

(+ A. V. Shokurov)



Uncertain reading. The surname after “A. B.” (A. V.) tentatively reads “Шокуров” (Shokurov).

Original source details: GELFAND-P000772

Landweber–Novikov algebras and formal vector fields on the line. [Crossed out: ...]



Undeciphered passage. Dense strokes obscure the small entry beneath the title.

Original source details: GELFAND-P000772

$\mathbb{R}^1$, Diff_1 — formal diffeomorphisms of the line with the additional condition: $\varphi'(0) = 1$

$$\varphi(0) = 0, \qquad \varphi'(0) = 1$$

The group $= \varprojlim$ finite-dimensional groups

(in jets of order n)

Lie algebra — the Lie algebra of formal vector fields on $\mathbb{R}^1$ — this is L_1.

$$t + a_1 t^2 + \ldots + a_n t^{n+1} + \ldots$$

a_j — coordinates.

Ring of functions: $\underline{P}$ (polynomials in all the a_j).



Uncertain reading. The ring is denoted by P with a lower stroke.

Original source details: GELFAND-P000772

Hirzebruch genus: M — a complex or stably complex manifold

[Crossed out: ...]



Undeciphered passage. A heavily crossed-out sign precedes M.

Original source details: GELFAND-P000772

$M \rightsquigarrow M(\varphi)$ — a polynomial on Diff_1.

$$\frac{t}{\varphi(t)} = \psi(t)$$

$$\prod_{j=1}^{N} \psi(\beta_j t), \qquad N > \dim_{\mathbb{C}} M$$

The coefficient of t^n = a symmetric function of β_j =

$= \Omega(\sigma_1, \ldots, \sigma_N)$, σ_j — elementary symmetric polynomials

GELFAND-P000773

[source number: 75]

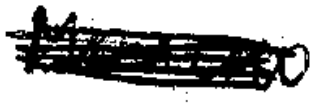
$\sigma_j \rightsquigarrow j$-th Chern class

Consider

$$\langle \Omega(\sigma_1, \ldots, \sigma_N), M \rangle$$

this is precisely the function on Diff_1.

[Crossed out: ...]



Uncertain reading. Dense strokes obscure several words before the line about operations.

Original source details: GELFAND-P000773

Operations on manifolds $\rightsquigarrow$ operations on polynomials.

left-invariant differential operators — ?

1) $M \times M_1$

2)

Diagram and typed description.

$$\begin{array}{c} \tau \\ \downarrow \\ M \end{array}$$



Original source details: GELFAND-P000773

— canonical bundle.

$c_1(\tau) \rightsquigarrow$ the cycle dual to it $M_1 \subset M$

$$\operatorname{Codim}_{\mathbb{C}} M_1 = 1.$$

(zeros of sections of the canonical bundle).

Analogue: dual to $\forall$ polynomial in the Chern classes.

Landweber–Novikov; there is nothing else. (these are all natural cohomological operations)

Theorem 1.

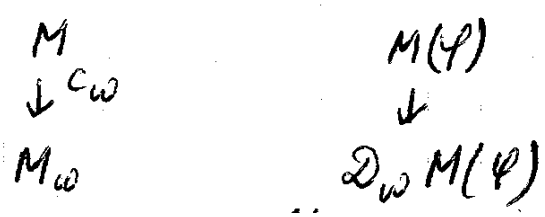


Diagram and typed description.

$$\begin{array}{ccc} M & & M(\varphi) \\ \downarrow{\scriptstyle c_\omega} & & \downarrow \\ M_\omega & & D_\omega M(\varphi) \end{array}$$

Original source details: GELFAND-P000773

c_ω — a polynomial in the Chern classes

D_ω — a left-invariant differential operator.

$\Longleftrightarrow$ a one-to-one correspondence

$M(\varphi)$ — integer polynomials

When do polynomials correspond to manifolds?

Answer:

Thm. 2.

$\mathrm{Diff}_1(\mathbb{Z})$ — formal diffeomorphisms with integer coordinates.

Todd genus $\varphi = 1 - \exp(-t)$ (diffeomorphism)

The polynomial must take integer values on the right coset of this φ [Crossed out: in?]



Uncertain reading. The heavily overwritten and cancelled letter tentatively reads “в” (in).

Original source details: GELFAND-P000773

$\mathrm{Diff}_1(\mathbb{Z})$.

[Crossed out: ...]



Uncertain reading. A separate crossed-out entry precedes the theorem reference.

Original source details: GELFAND-P000773

(∼ Thm. Stong–Hattori)

— · —

S — the set of all operations of Landweber–Novikov; this is an algebra.

Addition: disjoint union, multiplication — composition.

The cohomology of this algebra is important. All homotopy invariants of continuous maps are expressed through it. (Novikov)

Thm. 3

$$\mathrm{Ext}_S^*(\mathbb{Q},\mathbb{Q}) = H^*(L_1)$$

Cohomology of the algebra = cohomology of the enveloping algebra.

$H^*(L_1)$ was calculated in the works of Gelfand, Fuks, Goncharova

Problem: calculate the cohomology of the group $\mathrm{Diff}_1(\mathbb{Z})$.

GELFAND-P000774

Calogero. (and A. De gasperis).

A class of nonlinear evolution equations solvable by a spectral transform associated **with** Schrödinger.



Uncertain reading. The preposition is heavily retraced; it reads “c” (with).

Original source details: GELFAND-P000774

$$u_t(x,t) = \alpha(L,t)u_x(x,t) \tag{1}$$

$$Lf(x) = f_{xx} - 4u(x,t)f(x) + 2u_x(x,t)\int_x^\infty dx'\, f(x') \tag{2}$$

Sol. [Above the line, with an insertion mark: lin.] of the Cauchy problem reduces to a linear problem for the Fourier transform

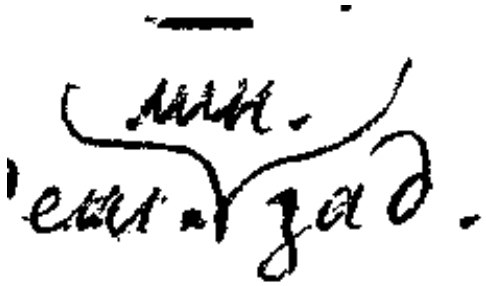


Diagram and typed description.

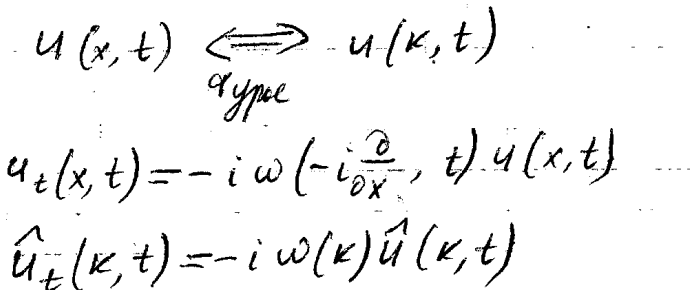
lin.
Sol.↓ problem

Original source details: GELFAND-P000774

$$u(x,t) \underset{\text{Fourier}}{\Longleftrightarrow} u(k,t)$$

$$u_t(x,t) = -i\omega\left(-i\frac{\partial}{\partial x}, t\right)u(x,t)$$

$$\hat{u}_t(k,t) = -i\omega(k)\hat{u}(k,t)$$



Original source details: GELFAND-P000774

Uncertain reading. The first right-hand $u(k,t)$ has no visible hat; the last line has $\omega(k)$ without the argument t.

A wider class:

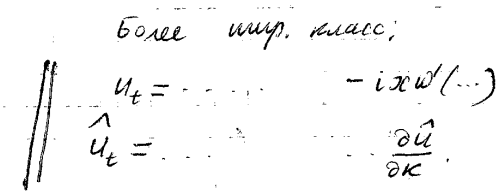


Diagram and typed description.

$$\begin{cases} u_t = \cdots - ix\omega'(\dots) \\ \hat{u}_t = \cdots \quad \cdots \dfrac{\partial \hat{u}}{\partial k} \end{cases}$$

Original source details: GELFAND-P000774

one can solve a wider class of equations:

$$\begin{aligned} u_t(x,t) = {} & \alpha(\underline{L},t)u_x(x,t) \\ & + \beta(\underline{L},t)[xu_x(x,t) + 2u(x,t)] \end{aligned} \tag{3}$$

α — a function of the operator

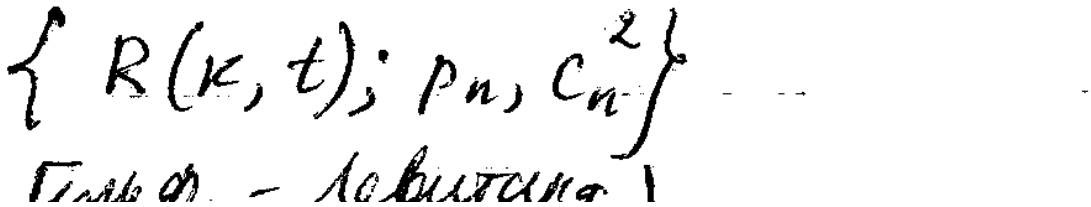
β — likewise

$$\alpha(\underline{L},t) = -\underline{L} \quad \Rightarrow \quad \text{KdV} \quad (\text{when } \beta = 0)$$

This includes the higher KdV equations: $\alpha(z,t) = z^k$.

For (3) there is a linear spectral problem describing the evolution of u.

$$u(x,t) \Longleftrightarrow \{R(k,t); p_n, c_n^2\}$$



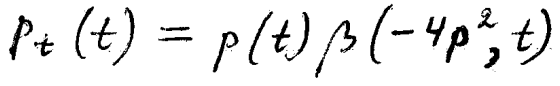
Original source details: GELFAND-P000774

Uncertain reading. p_n or ρ_n; the same sign recurs in (4').

(solution of the Gelfand–Levitan equation)

$$\begin{aligned} R_t(k,t) = {} & 2ik\alpha(-4k^2,t)R(k,t) \\ & - k\beta(-4k^2,t)R_k(k,t) \end{aligned} \tag{4}$$

$$p_t(t) = p(t)\beta(-4p^2,t) \tag{4'}$$



Uncertain reading. The shape transcribed as p at all three positions also permits the reading ρ.

Original source details: GELFAND-P000774

for c the equation is as for R (not written out)

The velocity of the solitons is not conserved, nor is the amplitude.

Conservation laws.

$$c^{(n)} = \int_{-\infty}^{\infty} dx\, \underline{L}^n[xu_x(x,t) + 2u(x,t)] \tag{5}$$

Theorem 1. $\beta \equiv 0 \quad \Rightarrow \quad (c^{(n)})_t = 0$
(in particular: KdV).

GELFAND-P000775

[source number: 76]

AKNS
(Ablowitz, Kaup, Newell, —)
— such a form of the higher KdV equations.
KdV is written for two Hamiltonian structures.
Poisson brackets — infinitely many.

A form of higher KdV with the integral: $\int_{-\infty}^{\infty} u^2 dx$.

(Faddeev et al. — the Leningrad group)
and a complicated Hamiltonian structure.

$$I_0 = \int u^2 dx;$$

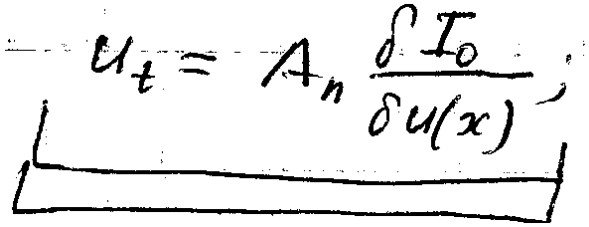


Diagram and typed description.

$$u_t = A_n \frac{\delta I_0}{\delta u(x)};$$

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these are the higher KdV equations

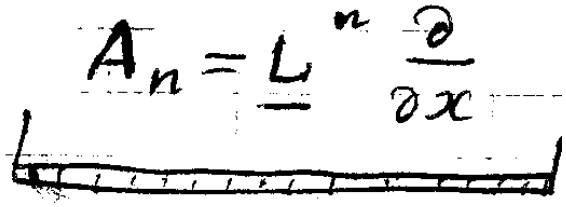


Diagram and typed description.

$$A_n = L^n \frac{\partial}{\partial x}$$

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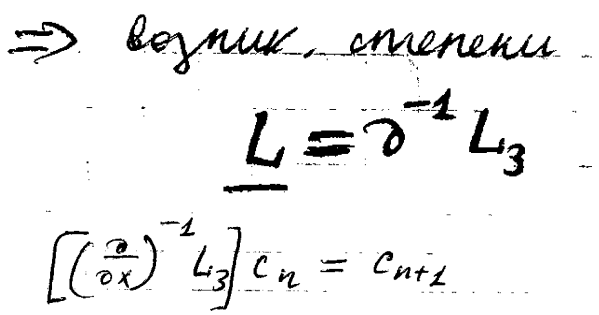
The transformation laws are invariant under shifts in x, and the integral contains x.

Novikov:

$$L_3 c_n = \frac{\partial}{\partial x} c_{n+1} \quad \Rightarrow \quad \left[\left(\frac{\partial}{\partial x}\right)^{-1} L_3\right] c_n = c_{n+1}$$

$\Rightarrow$ powers of the operator arise.

$$\underline{L} = \partial^{-1} L_3$$



Uncertain reading. In both entries the inverse derivative precedes L_3; the factor order is retained.

Original source details: GELFAND-P000775

Consider the Schrödinger equation.

$$\Psi_{xx} = [u(x) - k^2]\Psi \tag{6}$$

$$\Psi(k,x) \underset{x\to-\infty}{\longrightarrow} T(k)e^{-ikx}$$

$$\Psi(k,x) \underset{x\to+\infty}{\longrightarrow} e^{-ikx} + R(k)e^{ikx}$$

$$T(k) = |T(k)|e^{2i\varphi(k)}$$

$$T(\pm k) = |T(k)|e^{\pm 2i\varphi(k)}$$

$$\varphi(k) = \sum_{n=0}^{N} k^{-(2n+1)}\varphi^{(n)} + O(k^{-(2N+3)}). \tag{7}$$

Then

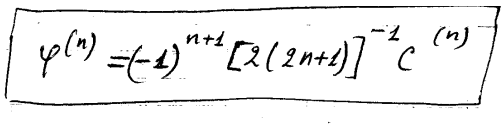


Diagram and typed description.

$$\varphi^{(n)} = (-1)^{n+1}[2(2n+1)]^{-1}c^{(n)} \tag{8}$$

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$$\begin{aligned} u_t(x,t) = {} & \alpha(\underline{L},t)u_x(x,t) \\ & + \beta(\underline{L},t)[xu_x(x,t) + 2u(x,t)] \end{aligned} \tag{9}$$

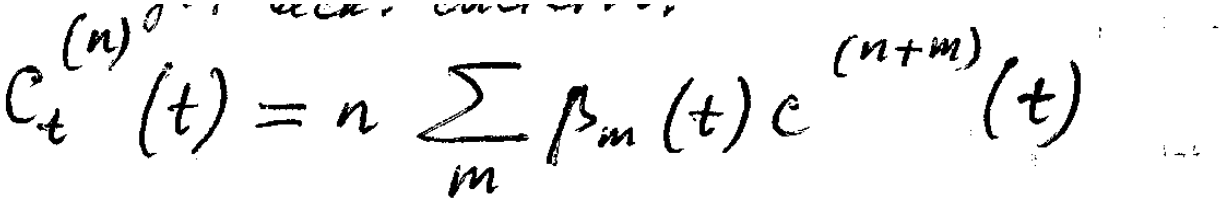
GELFAND-P000776

Let

$$\beta(z,t) = \sum \beta_n(t) z^n \tag{10}$$

Then one obtains an infinite system:

$$C_t^{(n)}(t) = n \sum_m \beta_m(t) c^{(n+m)}(t) \tag{11}$$



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Uncertain reading. The case of C/c in (11): a larger C on the left, a smaller c on the right.

β — a polynomial in $z \Rightarrow$ the system is triangular.

$\beta_0 = i\omega \Rightarrow$ all quantities are periodic in t.

(a consequence of the structure)

Next time — on the Bäcklund transform



Diagram and typed description. 21/III–78.

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V. P. Palamodov at the Society.

Lacunas, sharp fronts and deformations of singularities.

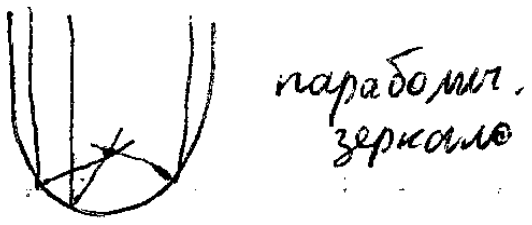
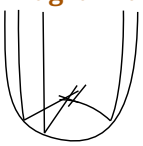


Diagram and typed description.



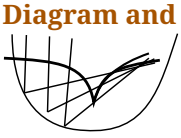
parabolic mirror

Short crossing strokes occur above the bottom; arrowheads cannot be distinguished.

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Diagram and typed description.



spherical mirror

Some crossings and junctions of the heavy lines cannot be distinguished.

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Caustic — “burning”, “hot” (in Greek)

Rays reflected from a spherical mirror are tangent to a curve with a singularity —
— the caustic.

Dynamic picture: wave fronts



Original source details: GELFAND-P000776

Diagram and typed description.



Some thin branches cross and merge with the heavy strokes; not every junction is established.

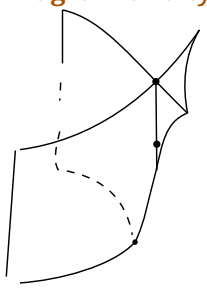
at the instant the caustic forms, a singularity $y = x^{4/3}$

There is a surface whose horizontal sections coincide with the positions of the front



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Diagram and typed description.



Gaps, crossings and possible overlaps of the surface are not always distinguishable; there are no labels.

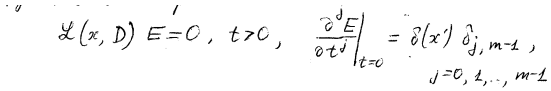
Consider a hyperbolic equation

$$\mathcal{L}(x, D)u = 0, \tag{1}$$

for example

$$\left(\frac{\partial^2}{\partial t^2} - \frac{1}{c^2}\Delta\right) u = 0 \tag{2}$$

Fundamental solution:



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Diagram and typed description.

$$\mathcal{L}(x, D)E = 0, \quad t > 0,$$
$$\left.\frac{\partial^j E}{\partial t^j}\right|_{t=0} = \delta(x')\delta_{j,m-1}, \qquad j = 0, 1, \ldots, m-1 \tag{3}$$